MF-Tyre/MF-Swift 6.2

Equation manual

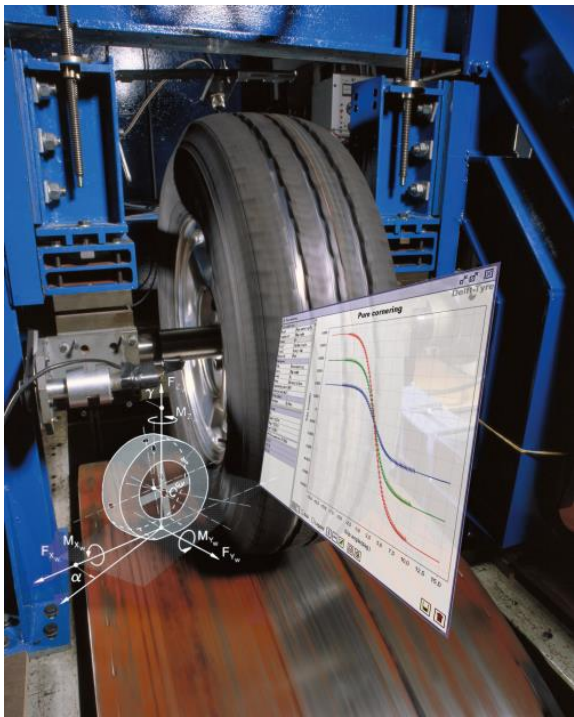

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Table of Contents

1 THE MAGIC FORMULA, STEADY-STATE BEHAVIOUR 4

1.1 INTRODUCTION 4
1.2 LONGITUDINAL FORCE FX 9
1.3 LATERAL FORCE FY 11
1.4 OVERTURNING MOMENT MX 16
1.5 ROLLING RESISTANCE MOMENT MY 17
1.6 SELF ALIGNING MOMENT MZ 18

2 BASIC TYRE PROPERTIES 21

2.1 LOADED RADIUS 21
2.2 EFFECTIVE ROLLING RADIUS 23
2.3 CONTACT PATCH DIMENSIONS 24
2.4 RELAXATION BEHAVIOUR 25

3 MF-SWIFT 29

3.1 CONTACT PATCH SLIP MODEL 29
3.2 RIGID RING 30
3.3 OBSTACLE ENVELOPING MODEL 33

4 MISCELLANEOUS 38

4.1 MOTORCYCLE CONTOUR ELLIPSE (MF-TYRE/MF-SWIFT 6.1 AND 6.2) 38
4.2 GYROSCOPIC MOMENT 38
4.3 VERTICAL STIFFNESS AND DAMPING TYRE 38
4.4 ROAD SURFACE SIMULATION SETTINGS 40

5 REFERENCES 41

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1 The Magic Formula, steady-state behaviour

1.1 Introduction

The Magic Formula is an empirical model to describe the non-linear tyre forces and moments occurring in the tyre contact patch for various rolling, sliding, loading conditions and tyre orientation. The Magic Formula can be considered as a nonlinear function with multiple inputs and outputs, see Figure 1.

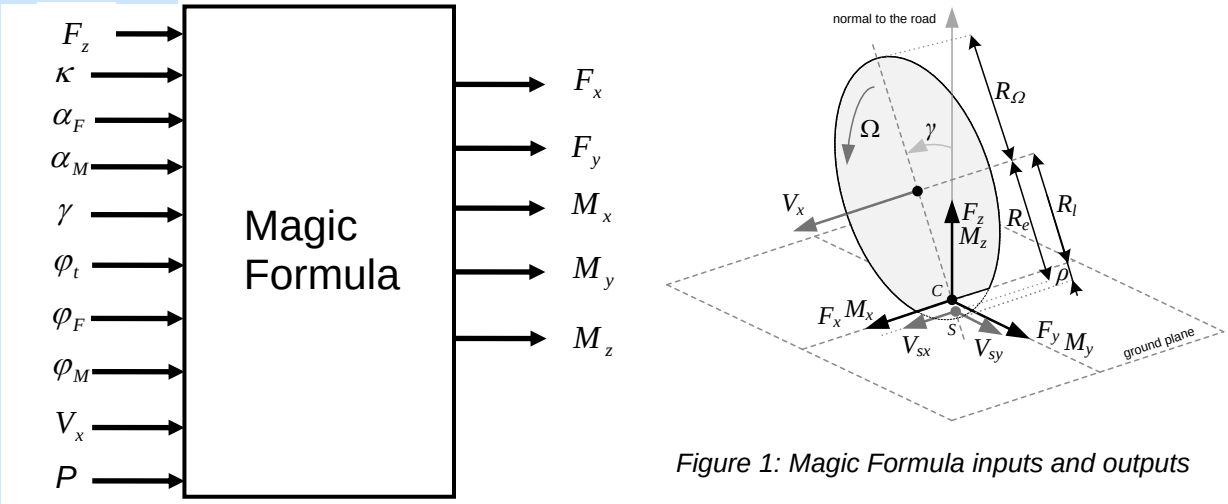

Figure 1: Magic Formula inputs and outputs

For steady-state rolling conditions the following input variables can be distinguished:

- vertical force F_z
- longitudinal slip $\kappa = -\frac{V_{sx}}{V_x} = -\frac{V_x - V_r}{V_x} = -\frac{V_x - \Omega R_e}{V_x}$
- side slip $\alpha_F = \arctan\left(\frac{V_{sy}}{|V_x|}\right),\ \alpha_M = \alpha_F$
- inclination angle γ
- turn slip $\varphi_t = -\frac{\dot{\psi}}{V}$
- spin $\varphi_F = -\frac{1}{V}\left(\dot{\psi} + (1-\varepsilon_\gamma)\Omega \sin\gamma\right),\ \varphi_M = \varphi_F$
- forward velocity V_x
- velocity $V = \sqrt{V_x^2 + V_{sy}^2}$

In addition the following variables are defined;

- longitudinal slip velocity V_{sx}
- lateral slip velocity V_{sy}
- wheel rotational velocity Ω
- effective rolling radius R_e (see also section 2.2)

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The following output variables are calculated:

- longitudinal force F_x
- lateral force F_y
- overturning moment M_x
- rolling resistance moment M_y
- self aligning moment M_z

Note: The difference between the side slip angles α_F and α_M vanishes for steady-state conditions, but becomes of importance for the dynamic tyre behaviour (the same applies for the spin quantities φ_F, φ_M).

In the Magic Formula various model parameters are made dimensionless using the unloaded tyre radius R_0, reference forward velocity V_{ref} and nominal vertical load F_{z0}. Starting with MF-Tyre 6.1 a nominal tyre pressure P_0 is introduced to make the parameters regarding tyre pressure dimensionless. The following equations hold:

load dependency

scaled nominal load: $F'_{z0} = \lambda_{Fz0} F_{z0}$

vertical load increment: $df_z = \dfrac{F_z - F'_{z0}}{F'_{z0}}$

inflation pressure dependency

pressure increment: $dp_i = \dfrac{P - P_0}{P_0}$

Note: The parameters $R_0, F_{z0}, V_{ref}, P_0$ should be fixed before performing the parameter identification and should never be changed afterwards!

In the turn slip equations a camber reduction factor ε_γ is defined.

$$\varepsilon_\gamma = p_{\varepsilon\gamma\varphi 1}\left(1 + p_{\varepsilon\gamma\varphi 2} df_z\right)$$

This reduction factor occurs in a number of expressions. Pacejka gives the following guidelines:

- motorcycle tyres: ε_γ is close to zero
- radial car tyres: ε_γ may reach values up to 0.7
- truck tyres ε_γ may become slightly over 1

<u>Limitations on the inputs of the Magic Formula</u>

The Magic Formula is an empirical tyre model which can accurately represent tyre measurements. When the inputs are far outside the measurement range (e.g. extremely high vertical loads or very large camber angles) the extrapolation capabilities of the model can possibly fail. To prevent this from happening the inputs to the Magic Formula are bounded and the Magic Formula is not evaluated outside this range. So:

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$$\kappa_{\min} < \kappa < \kappa_{\max}$$
$$\alpha_{\min} < \alpha_F < \alpha_{\max} \text{ and } \alpha_{\min} < \alpha_M < \alpha_{\max}$$
$$\gamma_{\min} < \gamma < \gamma_{\max}$$
$$P_{\min} < P < P_{\max}$$

For the vertical force also a range is defined, but additionally scaling of forces/moments is applied if the lower boundary is exceeded:

$F_z < F_{z,\min}$: Magic Formula evaluation for $F_{z,min}$ + linear scaling

$F_{z,\min} < F_z < F_{z,\max}$: Normal Magic Formula evaluation

$F_z > F_{z,\max}$: Magic Formula evaluation for $F_{z,max}$ + warning message

Simple example: the vertical force equals 0.5 times $F_{z,\min}$ then the Magic Formula is evaluated for $F_{z,\min}$ and the resulting forces/moments are multiplied with a factor 0.5.

Magic Formula versions

The first version of the Magic Formula was developed in 1987 [1], since then a number of new (and special) versions have been created.

In 1996 TNO introduced the commercial software MF-Tool for the Magic Formula parameter identification and tyre model MF-Tyre 5.0. These products were based on the equations presented in the 1996 paper of Pacejka [2] and implemented in a number of leading multibody simulation packages like ADAMS and DADS. Over the years some improvements were made (e.g. addition of the overturning moment, improved combined slip, etc.) and MF-Tyre 5.1 and MF-Tyre 5.2 were released consecutively. MF-Tyre was specifically intended for passenger car and truck tyres running at relatively small inclination angles (up to 10 degrees). For motorcycles running at much larger camber angles the Magic Formula equations had to be modified, this resulted in the tyre model MF-MCTyre 1.0 and an improved version MF-MCTyre 1.1 at later stage.

In MF-Tyre 6.0 the Magic Formula equations were modified, so that a single model can handle both car and motorcycle tyres. Also turn slip was added to represent for example the self-aligning moment occurring when twisting the tyre at standstill. Further, in MF-Tyre 6.1 the equations were extended to incorporate the effects of tyre inflation pressure changes. In MF-Tyre 6.2 the MF-Swift functionality is made available for motorcycle, further the loaded radius is upgraded for large side slip and camber angles.

The MF-Tyre/MF-Swift tyre software is backward compatible up to MF-Tyre 5.0. Since the equations have been modified over the years, the keyword FITTYP is present in the tyre property file to identify the Magic Formula version. The following convention applies:

FITTYP=5	MF-Tyre 5.0 and MF-Tyre 5.1
FITTYP=6	MF-Tyre 5.2
FITTYP=21	MF-Swift 1.x (based on MF-Tyre 5.2)
FITTYP=50	MF-MCTyre 1.0
FITTYP=51	MF-MCTyre 1.1
FITTYP=60	MF-Tyre/MF-Swift 6.0
FITTYP=61	MF-Tyre/MF-Swift 6.1
FITTYP=62	MF-Tyre/MF-Swift 6.2

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Note: Please do not change the value of FITTYP! FITTYP is used to select the appropriate Magic Formula equations. Once the parameters have been fitted with a specific FITTYP you should never change its value afterwards.

In the next sections the Magic Formula equations are presented. The description covers MF-Tyre 5.2, MF-MCTyre 1.1, MF-Tyre 6.0, MF-Tyre 6.1 and MF-Tyre 6.2.

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	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[MODEL]										
	FITTYP	Magic Formula version number	62	62	61	61	60	60	6	21	52
	TYRESIDE	Position of tyre during measurements	x	x	x	x	x	x	x	x	x
V_{ref}	LONGVL	Reference speed	x	x	x	x	x	x	x	x	x
	[DIMENSION]										
R_0	UNLOADED_RADIUS	Free tyre radius	x	x	x	x	x	x	x	x	x
	[OPERATING_CONDITIONS]										
P	INFLPRES	Tyre inflation pressure	x	x	x	x					
P_0	NOMPRES	Nominal pressure used in (MF) equations	x	x	x	x					
	[VERTICAL]										
F_{z0}	FNOMIN	Nominal wheel load	x	x	x	x	x	x	x	x	x
	[INFLATION_PRESSURE_RANGE]										
$P_{\min}$	PRESMIN	Minimum allowed inflation pressure	x	x	x	x					
$P_{\max}$	PRESMAX	Maximum allowed inflation pressure	x	x	x	x					
	[VERTICAL_FORCE_RANGE]										
$F_{z,\min}$	FZMIN	Minimum allowed wheel load	x	x	x	x	x	x	x	x	x
$F_{z,\max}$	FZMAX	Maximum allowed wheel load	x	x	x	x	x	x	x	x	x
	[LONG_SLIP_RANGE]										
$\kappa_{\min}$	KPUMIN	Minimum valid wheel slip	x	x	x	x	x	x	x	x	x
$\kappa_{\max}$	KPUMAX	Maximum valid wheel slip	x	x	x	x	x	x	x	x	x
	[SLIP_ANGLE_RANGE]										
$\alpha_{\min}$	ALPMIN	Minimum valid slip angle	x	x	x	x	x	x	x	x	x
$\alpha_{\max}$	ALPMAX	Maximum valid slip angle	x	x	x	x	x	x	x	x	x
	[INCLINATION_ANGLE_RANGE]										
$\gamma_{\min}$	CAMMIN	Minimum valid camber angle	x	x	x	x	x	x	x	x	x
$\gamma_{\max}$	CAMMAX	Maximum valid camber angle	x	x	x	x	x	x	x	x	x
	[SCALING_COEFFICIENTS]										
λ_{Fz0}	LFZO	Scale factor of nominal (rated) load	x	x	x	x	x	x	x	x	x
	[TURNSLIP_COEFFICIENTS]										
$p_{\varepsilon\gamma\varphi 1}$	PECP1	Camber w.r.t. spin reduction factor parameter in camber stiffness	x	x	x	x	x	x			
$p_{\varepsilon\gamma\varphi 2}$	PECP2	Camber w.r.t. spin reduction factor varying with load parameter in camber stiffness	x	x	x	x	x	x			

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1.2 Longitudinal force Fx

$$F_x = \left(D_x \sin\left[C_x \arctan\left\{B_x\kappa_x - E_x\left(B_x\kappa_x - \arctan\left(B_x\kappa_x\right)\right)\right\}\right] + S_{Vx}\right)\cdot G_{x\alpha}$$

Pure slip:

$$\kappa_x = \kappa + S_{Hx}$$

$$C_x = p_{Cx1}\cdot\lambda_{Cx}$$

$$D_x = \mu_x \cdot F_z \cdot \zeta_1$$

$$\mu_x = \left(p_{Dx1} + p_{Dx2}df_z\right)\cdot\left(1 + p_{px3}dpi + p_{px4}dpi^2\right)\cdot\left(1 - p_{Dx3}\gamma^2\right)\cdot\lambda_{\mu x}$$

$$E_x = \left(p_{Ex1} + p_{Ex2}df_z + p_{Ex3}df_z^2\right)\cdot\left(1 - p_{Ex4}\,\mathrm{sgn}\left(\kappa_x\right)\right)\cdot\lambda_{Ex} \le 1$$

$$K_{x\kappa} = F_z\cdot\left(p_{Kx1} + p_{Kx2}df_z\right)\cdot\exp\left(p_{Kx3}df_z\right)\cdot\left(1 + p_{px1}dpi + p_{px2}dpi^2\right)\cdot\lambda_{Kx\kappa}$$

$$B_x = \frac{K_{x\kappa}}{C_x D_x}$$

MF-Tyre 5.2, 6.0, 6.1, 6.2: $S_{Hx} = \left(p_{Hx1} + p_{Hx2}df_z\right)\cdot\lambda_{Hx}$

MF-MCTyre 1.1: $S_{Hx} = -\dfrac{\left(q_{sy1}F_z\lambda_{My} + S_{Vx}\right)}{K_{x\kappa}}$

$$S_{Vx} = F_z\cdot\left(p_{Vx1} + p_{Vx2}df_z\right)\cdot\lambda_{Vx}\cdot\lambda_{\mu x}\cdot\zeta_1$$

Combined slip:

$$G_{x\alpha} = \frac{\cos\left[C_{x\alpha}\arctan\left\{B_{x\alpha}\alpha_s - E_{x\alpha}\left(B_{x\alpha}\alpha_s - \arctan\left(B_{x\alpha}\alpha_s\right)\right)\right\}\right]}{\cos\left[C_{x\alpha}\arctan\left\{B_{x\alpha}S_{Hx\alpha} - E_{x\alpha}\left(B_{x\alpha}S_{Hx\alpha} - \arctan\left(B_{x\alpha}S_{Hx\alpha}\right)\right)\right\}\right]}$$

$$\alpha_s = \alpha_F + S_{Hx\alpha}$$

$$B_{x\alpha} = (r_{Bx1} + r_{Bx3}\gamma^2)\cos\{\arctan[r_{Bx2}\kappa]\}\lambda_{xa}$$

$$C_{x\alpha} = r_{Cx1}$$

$$E_{x\alpha} = r_{Ex1} + r_{Ex2}df_z$$

$$S_{Hx\alpha} = r_{Hx1}$$

when combined slip is not used: $G_{x\alpha} = 1$

Turn slip:

$$\zeta_1 = \cos\left(\arctan\left(B_{x\varphi}R_0\varphi_F\right)\right)$$

$$B_{x\varphi} = p_{Dx\varphi1}\left(1 + p_{Dx\varphi2}df_z\right)\cos\left(\arctan\left(p_{Dx\varphi3}\kappa\right)\right)$$

when turn slip is not used: $\zeta_1 = 1$

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Tyre property file

			MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[SCALING_COEFFICIENTS]										
λ_{Cx}	LCX	Scale factor of Fx shape factor	x	x	x	x	x	x	x	x	x
$\lambda_{\mu x}$	LMUX	Scale factor of Fx peak friction coefficient	x	x	x	x	x	x	x	x	x
λ_{Ex}	LEX	Scale factor of Fx curvature factor	x	x	x	x	x	x	x	x	x
$\lambda_{Kx\kappa}$	LKX	Scale factor of slip stiffness	x	x	x	x	x	x	x	x	x
λ_{Hx}	LHX	Scale factor of Fx horizontal shift	x	x	x	x	x	x	x	x	
λ_{Vx}	LVX	Scale factor of Fx vertical shift	x	x	x	x	x	x	x	x	x
$\lambda_{x\alpha}$	LXAL	Scale factor of alpha influence on Fx	x	x	x	x	x	x	x	x	x
	[LONGITUDINAL_COEFFICIENTS]										
p_{Cx1}	PCX1	Shape factor Cfx for longitudinal force	x	x	x	x	x	x	x	x	x
p_{Dx1}	PDX1	Longitudinal friction Mux at Fznom	x	x	x	x	x	x	x	x	x
p_{Dx2}	PDX2	Variation of friction Mux with load	x	x	x	x	x	x	x	x	x
p_{Dx3}	PDX3	Variation of friction Mux with camber	x	x	x	x	x	x	x	x	x
p_{Ex1}	PEX1	Longitudinal curvature Efx at Fznom	x	x	x	x	x	x	x	x	x
p_{Ex2}	PEX2	Variation of curvature Efx with load	x	x	x	x	x	x	x	x	x
p_{Ex3}	PEX3	Variation of curvature Efx with load squared	x	x	x	x	x	x	x	x	x
p_{Ex4}	PEX4	Factor in curvature Efx while driving	x	x	x	x	x	x	x	x	x
p_{Kx1}	PKX1	Longitudinal slip stiffness Kfx/Fz at Fznom	x	x	x	x	x	x	x	x	x
p_{Kx2}	PKX2	Variation of slip stiffness Kfx/Fz with load	x	x	x	x	x	x	x	x	x
p_{Kx3}	PKX3	Exponent in slip stiffness Kfx/Fz with load	x	x	x	x	x	x	x	x	x
p_{Hx1}	PHX1	Horizontal shift Shx at Fznom	x	x	x	x	x	x	x	x	
p_{Hx2}	PHX2	Variation of shift Shx with load	x	x	x	x	x	x	x	x	
p_{Vx1}	PVX1	Vertical shift Svx/Fz at Fznom	x	x	x	x	x	x	x	x	x
p_{Vx2}	PVX2	Variation of shift Svx/Fz with load	x	x	x	x	x	x	x	x	x
r_{Bx1}	RBX1	Slope factor for combined slip Fx reduction	x	x	x	x	x	x	x	x	x
r_{Bx2}	RBX2	Variation of slope Fx reduction with kappa	x	x	x	x	x	x	x	x	x
r_{Bx3}	RBX3	Influence of camber on stiffness for Fx combined	x	x	x	x	x	x			x
r_{Cx1}	RCX1	Shape factor for combined slip Fx reduction	x	x	x	x	x	x	x	x	x
r_{Ex1}	REX1	Curvature factor of combined Fx	x	x	x	x	x	x	x	x	x
r_{Ex2}	REX2	Curvature factor of combined Fx with load	x	x	x	x	x	x	x	x	x
r_{Hx1}	RHX1	Shift factor for combined slip Fx reduction	x	x	x	x	x	x	x	x	x
p_{px1}	PPX1	Linear pressure effect on slip stiffness	x	x	x	x					
p_{px2}	PPX2	Quadratic pressure effect on slip stiffness	x	x	x	x					
p_{px3}	PPX3	Linear pressure effect on longitudinal friction	x	x	x	x					
p_{px4}	PPX4	Quadratic pressure effect on longitudinal friction	x	x	x	x					
	[TURNSLIP_COEFFICIENTS]										
$p_{Dx\varphi 1}$	PDXP1	Peak Fx reduction due to spin parameter	x	x	x	x	x	x			
$p_{Dx\varphi 2}$	PDXP2	Peak Fx reduction due to spin with varying load parameter	x	x	x	x	x	x			
$p_{Dx\varphi 3}$	PDXP3	Peak Fx reduction due to spin with kappa parameter	x	x	x	x	x	x			

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1.3 Lateral force Fy

$$F_y = G_{y\kappa} F_{yp} + S_{Vy\kappa}$$

Pure slip: MF-Tyre

$$F_{yp} = D_y \sin\left[C_y \arctan\left\{B_y \alpha_y - E_y\left(B_y \alpha_y - \arctan\left(B_y \alpha_y\right)\right)\right\}\right] + S_{Vy}$$

$$\alpha_y = \alpha_F + S_{Hy}$$

$$C_y = p_{Cy1} \lambda_{Cy}$$

$$D_y = \mu_y \cdot F_z \cdot \zeta_2$$

$$\mu_y = \left(p_{Dy1} + p_{Dy2} df_z\right) \cdot \left(1 + p_{py3} dpi + p_{py4} dpi^2\right) \cdot \left(1 - p_{Dy3} \gamma^2\right) \cdot \lambda_{\mu y}$$

$$E_y = \left(p_{Ey1} + p_{Ey2} df_z\right) \cdot \left(1 + p_{Ey5} \gamma^2 - \left(p_{Ey3} + p_{Ey4} \gamma\right) \operatorname{sgn}\left(\alpha_y\right)\right) \lambda_{Ey} \leq 1$$

$$K_{y\alpha} = p_{Ky1} \cdot F_{z0}' \cdot \left(1 + p_{py1} dpi\right) \cdot \sin\left[p_{Ky4} \arctan\left\{\frac{F_z}{\left(p_{Ky2} + p_{Ky5} \gamma^2\right) F_{z0}' \cdot \left(1 + p_{py2} dpi\right)}\right\}\right] \cdot \left(1 - p_{Ky3} |\gamma|\right) \cdot \lambda_{Ky\alpha} \cdot \zeta_3$$

$$K_{y\alpha 0} = p_{Ky1} \cdot F_{z0}' \cdot \left(1 + p_{py1} dpi\right) \cdot \sin\left[p_{Ky4} \arctan\left\{\frac{F_z}{p_{Ky2} F_{z0}' \cdot \left(1 + p_{py2} dpi\right)}\right\}\right] \cdot \lambda_{Ky\alpha}$$

MF-Tyre 5.2: $K_{y\gamma 0} = \left(p_{Hy3} K_{y\alpha 0} + F_z \left(p_{Vy3} + p_{Vy4} df_z\right)\right) \lambda_{Ky\gamma}$

MF-Tyre 6.0: $K_{y\gamma 0} = \left(p_{Ky6} + p_{Ky7} df_z\right) F_z \lambda_{Ky\gamma}$

MF-Tyre 6.1, 6.2: $K_{y\gamma 0} = \left(p_{Ky6} + p_{Ky7} df_z\right) F_z \lambda_{Ky\gamma} \cdot \left(1 + p_{py5} dpi\right)$

$$B_y = \frac{K_{y\alpha}}{C_y D_y}$$

$$S_{Hy0} = \left(p_{Hy1} + p_{Hy2} df_z\right) \cdot \lambda_{Hy}$$

MF-Tyre 5.2: $S_{Hy\gamma} = p_{Hy3} \cdot \gamma \cdot \lambda_{Ky\gamma}$

MF-Tyre 6.0, 6.1, 6.2: $S_{Hy\gamma} = \frac{K_{y\gamma 0} \gamma - S_{Vy\gamma}}{K_{y\alpha}} \zeta_0 + \zeta_4 - 1$

$$S_{Hy} = S_{Hy0} + S_{Hy\gamma}$$

$$S_{Vy0} = F_z \left(p_{Vy1} + p_{Vy2} df_z\right) \cdot \lambda_{Vy} \lambda_{\mu y}$$

$$S_{Vy\gamma} = F_z \left(p_{Vy3} + p_{Vy4} df_z\right) \cdot \gamma \cdot \lambda_{Ky\gamma} \lambda_{\mu y} \zeta_2$$

$$S_{Vy} = S_{Vy0} \zeta_2 + S_{Vy\gamma}$$

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Pure slip: MF-MC-Tyre 1.1

$$F_{yp} = D_y \sin\begin{bmatrix} C_y \arctan\{B_y\alpha_y - E_y(B_y\alpha_y - \arctan(B_y\alpha_y))\} \\ + C_\gamma \arctan\{B_\gamma\gamma - E_\gamma(B_\gamma\gamma - \arctan(B_\gamma\gamma))\} \end{bmatrix} + S_{Vy}$$

$$\alpha_y = \alpha_F + S_{Hy}$$

$$C_y = p_{Cy1}\lambda_{Cy}$$

$$C_\gamma = p_{Cy2}\lambda_{C\gamma}$$

$$D_y = \mu_y \cdot F_z$$

$$\mu_y = p_{Dy1} \cdot \exp(p_{Dy2}df_z) \cdot (1 + p_{py3}dpi + p_{py4}dpi^2) \cdot \lambda_{\mu y}(1 - p_{Dy3}\gamma^2)$$

$$K_{y\alpha} = p_{Ky1} \cdot F_{z0}' \cdot (1 + p_{py1}dpi) \cdot \sin\left[p_{Ky2}\arctan\left\{\frac{F_z}{(p_{Ky3} + p_{Ky4}\gamma^2)F_{z0}' \cdot (1 + p_{py2}dpi)}\right\}\right] \cdot (1 - p_{Ky5}\gamma^2) \cdot \lambda_{Ky\alpha} \cdot \zeta_3$$

$$K_{y\alpha 0} = p_{Ky1} \cdot F_{z0}' \cdot (1 + p_{py1}dpi) \cdot \sin\left[p_{Ky2}\arctan\left\{\frac{F_z}{P_{Ky3}F_{z0}' \cdot (1 + p_{py2}dpi)}\right\}\right] \cdot \lambda_{Ky\alpha}$$

$$K_{y\gamma 0} = (p_{Ky6} + p_{Ky7}df_z)F_z\lambda_{Ky\gamma} \cdot (1 + p_{py5}dpi)$$

$$B_y = \frac{K_{y\alpha}}{C_y D_y}$$

$$B_\gamma = \frac{K_{y\gamma 0}}{C_\gamma D_y}$$

$$E_y = \{p_{Ey1} + p_{Ey2}\gamma^2 + (p_{Ey3} + p_{Ey4}\gamma)\operatorname{sgn}(\alpha_y)\}\lambda_{Ey}$$

$$E_\gamma = p_{Ey5}\lambda_{E\gamma}$$

$$S_{Hy} = (p_{Hy1} + P_{Hy2}df_z) \cdot \lambda_{Hy}$$

$$S_{Vy} = F_z(p_{Vy1} + P_{Vy2}df_z) \cdot \lambda_{Vy}\lambda_{\mu y}\zeta_2$$

$$S_{Vy\gamma} = F_z(p_{Vy3} + p_{Vy4}df_z) \cdot \gamma \cdot \lambda_{Ky\gamma}\lambda_{\mu y}\zeta_2$$

Combined slip (all models):

braking induced plysteer

$$D_{Vy\kappa} = \mu_y F_z(r_{Vy1} + r_{Vy2}df_z + r_{Vy3}\gamma)\cos(\arctan(r_{Vy4}\alpha_F)) \cdot \zeta_2$$

$$S_{Vy\kappa} = D_{Vy\kappa}\sin(r_{Vy5}\arctan(r_{Vy6}\kappa))\lambda_{Vy\kappa}$$

weighting function:

$$G_{y\kappa} = \frac{\cos[C_{y\kappa}\arctan\{B_{y\kappa}\kappa_s - E_{y\kappa}(B_{y\kappa}\kappa_s - \arctan(B_{y\kappa}\kappa_s))\}]}{\cos[C_{y\kappa}\arctan\{B_{y\kappa}S_{Hy\kappa} - E_{y\kappa}(B_{y\kappa}S_{Hy\kappa} - \arctan(B_{y\kappa}S_{Hy\kappa}))\}]}$$

$$\kappa_s = \kappa + S_{Hy\kappa}$$

$$B_{y\kappa} = (r_{By1} + r_{By4}\gamma^2)\cos\{\arctan[r_{By2}(\alpha_F - r_{By3})]\}\lambda_{y\kappa}$$

$$C_{y\kappa} = r_{Cy1}$$

$$E_{y\kappa} = r_{Ey1} + r_{Ey2}df_z$$

$$S_{Hy\kappa} = r_{Hy1} + r_{Hy2}df_z$$

when combined slip is not used: $S_{Vy\kappa} = 0$, $G_{y\kappa} = 1$

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Turn slip:

$$\zeta_0 = 0$$

$$\zeta_2 = \cos\left(\arctan\left(B_{y\varphi}\left(R_0|\varphi_F| + p_{Dy\varphi 4}\sqrt{R_0|\varphi_F|}\right)\right)\right)$$

$$B_{y\varphi} = p_{Dy\varphi 1}\left(1 + p_{Dy\varphi 2} df_z\right)\cos\left(\arctan\left(p_{Dy\varphi 3}\tan\alpha_F\right)\right)$$

$$\zeta_3 = \cos\left(\arctan\left(p_{Ky\varphi 1} R_0{}^2 \varphi_F{}^2\right)\right)$$

$$\zeta_4 = 1 + S_{Hy\varphi} - S_{Vy\gamma}/K_{y\alpha}$$

$$S_{Hy\varphi} = D_{Hy\varphi}\sin\left(C_{Hy\varphi}\arctan\left(B_{Hy\varphi}R_0\varphi_F - E_{Hy\varphi}\left(B_{Hy\varphi}R_0\varphi_F - \arctan\left(B_{Hy\varphi}R_0\varphi_F\right)\right)\right)\right)$$

$$C_{Hy\varphi} = p_{Hy\varphi 1}$$

$$D_{Hy\varphi} = p_{Hy\varphi 2} + p_{Hy\varphi 3} df_z$$

$$E_{Hy\varphi} = p_{Hy\varphi 4}$$

$$K_{yR\varphi 0} = \frac{K_{y\gamma 0}}{1 - \varepsilon_\gamma}$$

$$B_{Hy\varphi} = -K_{yR\varphi 0}/\left(C_{Hy\varphi}D_{Hy\varphi}K_{y\alpha 0}\right)$$

when turn slip is not used: $\zeta_0, \zeta_2, \zeta_3, \zeta_4 = 1$

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	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[SCALING_COEFFICIENTS]										
λ_{Cy}	LCY	Scale factor of Fy shape factor	x	x	x	x	x	x	x	x	x
$\lambda_{\mu y}$	LMUY	Scale factor of Fy peak friction coefficient	x	x	x	x	x	x	x	x	x
λ_{Ey}	LEY	Scale factor of Fy curvature factor	x	x	x	x	x	x	x	x	x
$\lambda_{Ky\alpha}$	LKY	Scale factor of cornering stiffness	x	x	x	x	x	x	x	x	x
$\lambda_{Ky\gamma}$	LKYC	Scale factor of camber stiffness	x	x	x	x	x	x			
$\lambda_{Kz\gamma}$	LKZC	Scale factor of camber moment stiffness	x	x	x	x	x	x			
λ_{Hy}	LHY	Scale factor of Fy horizontal shift	x	x	x	x	x	x	x	x	x
λ_{Vy}	LVY	Scale factor of Fy vertical shift	x	x	x	x	x	x	x	x	
$\lambda_{y\kappa}$	LYKA	Scale factor of kappa influence on Fy	x	x	x	x	x	x	x	x	x
$\lambda_{Vy\kappa}$	LVYKA	Scale factor of kappa induced 'plysteer' Fy	x	x	x	x	x	x	x	x	x
$\lambda_{Ky\gamma}$	LKC	Scale factor of camber stiffness									x
$\lambda_{C\gamma}$	LCC	Scale factor of camber shape factor									x
$\lambda_{E\gamma}$	LEC	Scale factor of camber curvature factor									x
	[LATERAL_COEFFICIENTS]										
p_{Cy1}	PCY1	Shape factor Cfy for lateral forces	x	x	x	x	x	x	x	x	x
p_{Cy2}	PCY2	Shape factor Cfc for camber forces									x
p_{Dy1}	PDY1	Lateral friction Muy	x	x	x	x	x	x	x	x	x
p_{Dy2}	PDY2	Variation of friction Muy with load	x	x	x	x	x	x	x	x	x
p_{Dy3}	PDY3	Variation of friction Muy with squared camber	x	x	x	x	x	x	x	x	x
p_{Ey1}	PEY1	Lateral curvature Efy at Fznom	x	x	x	x	x	x	x	x	x
p_{Ey2}	PEY2	Variation of curvature Efy with load	x	x	x	x	x	x	x	x	x
p_{Ey3}	PEY3	Zero order camber dependency of curvature Efy	x	x	x	x	x	x	x	x	x
p_{Ey4}	PEY4	Variation of curvature Efy with camber	x	x	x	x	x	x	x	x	x
p_{Ey5}	PEY5	Camber curvature Efc	x	x	x	x	x	x			x
p_{Ky1}	PKY1	Maximum value of stiffness Kfy/Fznom	x	x	x	x	x	x	x	x	x
p_{Ky2}	PKY2	Load at which Kfy reaches maximum value	x	x	x	x	x	x	x	x	x
p_{Ky3}	PKY3	Variation of Kfy/Fznom with camber	x	x	x	x	x	x	x	x	x
p_{Ky4}	PKY4	Peak stiffness variation with camber squared	x	x	x	x	x	x			x
p_{Ky5}	PKY5	Lateral stiffness depedency with camber	x	x	x	x	x	x			x
p_{Ky6}	PKY6	Camber stiffness factor	x	x	x	x	x	x			x
p_{Ky7}	PKY7	Load dependency of camber stiffness factor	x	x	x	x	x	x			x
p_{Hy1}	PHY1	Horizontal shift Shy at Fznom	x	x	x	x	x	x	x	x	x
p_{Hy2}	PHY2	Variation of shift Shy with load	x	x	x	x	x	x	x	x	
p_{Hy3}	PHY3	Variation of shift Shy with camber							x	x	
p_{Vy1}	PVY1	Vertical shift in Svy/Fz at Fznom	x	x	x	x	x	x	x	x	
p_{Vy2}	PVY2	Variation of shift Svy/Fz with load	x	x	x	x	x	x	x	x	
p_{Vy3}	PVY3	Variation of shift Svy/Fz with camber	x	x	x	x	x	x	x	x	
p_{Vy4}	PVY4	Variation of shift Svy/Fz with camber and load	x	x	x	x	x	x	x	x	

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Tyre property file

			MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[LATERAL_COEFFICIENTS]										
r_{By1}	RBY1	Slope factor for combined Fy reduction	x	x	x	x	x	x	x	x	x
r_{By2}	RBY2	Variation of slope Fy reduction with alpha	x	x	x	x	x	x	x	x	x
r_{By3}	RBY3	Shift term for alpha in slope Fy reduction	x	x	x	x	x	x	x	x	x
r_{By4}	RBY4	Influence of camber on stiffness of Fy combined	x	x	x	x	x	x			x
r_{Cy1}	RCY1	Shape factor for combined Fy reduction	x	x	x	x	x	x	x	x	x
r_{Ey1}	REY1	Curvature factor of combined Fy	x	x	x	x	x	x	x	x	x
r_{Ey2}	REY2	Curvature factor of combined Fy with load	x	x	x	x	x	x	x	x	x
r_{Hy1}	RHY1	Shift factor for combined Fy reduction	x	x	x	x	x	x	x	x	x
r_{Hy2}	RHY2	Shift factor for combined Fy reduction with load	x	x	x	x	x	x	x	x	x
r_{Vy1}	RVY1	Kappa induced side force Svyk/Muy*Fz at Fznom	x	x	x	x	x	x	x	x	x
r_{Vy2}	RVY2	Variation of Svyk/Muy*Fz with load	x	x	x	x	x	x	x	x	x
r_{Vy3}	RVY3	Variation of Svyk/Muy*Fz with camber	x	x	x	x	x	x	x	x	x
r_{Vy4}	RVY4	Variation of Svyk/Muy*Fz with alpha	x	x	x	x	x	x	x	x	x
r_{Vy5}	RVY5	Variation of Svyk/Muy*Fz with kappa	x	x	x	x	x	x	x	x	x
r_{Vy6}	RVY6	Variation of Svyk/Muy*Fz with atan(kappa)	x	x	x	x	x	x	x	x	x
p_{py1}	PPY1	Pressure effect on cornering stiffness magnitude	x	x	x	x					
p_{py2}	PPY2	Pressure effect on location of cornering stiffness peak	x	x	x	x					
p_{py3}	PPY3	Linear pressure effect on lateral friction	x	x	x	x					
p_{py4}	PPY4	Quadratic pressure effect on lateral friction	x	x	x	x					
p_{py5}	PPY5	Influence of inflation pressure on camber stiffness	x	x	x	x					
	[TURNSLIP_COEFFICIENTS]										
$p_{Ky\varphi 1}$	PKYP1	Cornering stiffness reduction due to spin	x	x	x	x	x	x			
$p_{Dy\varphi 1}$	PDYP1	Peak Fy reduction due to spin parameter	x	x	x	x	x	x			
$p_{Dy\varphi 2}$	PDYP2	Peak Fy reduction due to spin with varying load parameter	x	x	x	x	x	x			
$p_{Dy\varphi 3}$	PDYP3	Peak Fy reduction due to spin with alpha parameter	x	x	x	x	x	x			
$p_{Dy\varphi 4}$	PDYP4	Peak Fy reduction due to square root of spin parameter	x	x	x	x	x	x			
$p_{Hy\varphi 1}$	PHYP1	Fy-alpha curve lateral shift limitation	x	x	x	x	x	x			
$p_{Hy\varphi 2}$	PHYP2	Fy-alpha curve maximum lateral shift parameter	x	x	x	x	x	x			
$p_{Hy\varphi 3}$	PHYP3	Fy-alpha curve maximum lateral shift varying with load parameter	x	x	x	x	x	x			
$p_{Hy\varphi 4}$	PHYP4	Fy-alpha curve maximum lateral shift parameter	x	x	x	x	x	x			

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1.4 Overturning moment Mx

$$M_x = R_0 F_z \lambda_{Mx} \Bigg\{ q_{sx1}\lambda_{VMx} - q_{sx2}\gamma\left(1 + p_{pmx1} dpi\right) + q_{sx3}\frac{F_y}{F_{z0}}$$

$$+ q_{sx4}\cos\left[q_{sx5}\arctan\left(\left(q_{sx6}\frac{F_z}{F_{z0}}\right)^2\right)\right]\cdot\sin\left[q_{sx7}\gamma + q_{sx8}\arctan\left(q_{sx9}\frac{F_y}{F_{z0}}\right)\right]$$

$$+ q_{sx10}\arctan\left(q_{sx11}\frac{F_z}{F_{z0}}\right)\gamma\Bigg\} + \underbrace{R_0\lambda_{Mx}\left(F_y\left\{q_{sx13} + q_{sx14}|\gamma|\right\} - F_z q_{sx12}\gamma|\gamma|\right)}_{alternative\ equation}$$

Important note:

The formula for M_x consists of two separate equations that are separated with the "+" sign. The default equation is the first part with coefficients q_{sx1}, q_{sx2}, … , q_{sx11}. When using this formulation q_{sx12}, q_{sx13} and q_{sx14} must be equal to zero! The second part contains an alternative formulation with coefficients q_{sx12}, q_{sx13} and q_{sx14} and might be used for special motorcycle tyres. When using this part coefficients q_{sx1}, q_{sx2}, … , q_{sx11} must be set to zero. Combinations of both equation parts lead to erroneous results and should consequently not be applied.

	Tyre property file		**MF-Tyre 6.2**	**MF-Swift 6.2**	**MF-Tyre 6.1**	**MF-Swift 6.1**	**MF-Tyre 6.0**	**MF-Swift 6.0**	**MF-Tyre 5.2**	**SWIFT 1.2**	**MF-MCTyre 1.1**
	[SCALING_COEFFICIENTS]										
λ_{Mx}	LMX	Scale factor of overturning moment	x	x	x	x	x	x	x	x	x
λ_{VMx}	LVMX	Scale factor of Mx vertical shift	x	x	x	x	x	x	x	x	x
	[OVERTURNING_COEFFICIENTS]										
q_{sx1}	QSX1	Overturning moment offset	x	x	x	x	x	x	x	x	x
q_{sx2}	QSX2	Camber induced overturning couple	x	x	x	x	x	x	x	x	x
q_{sx3}	QSX3	Fy induced overturning couple	x	x	x	x	x	x	x	x	x
q_{sx4}	QSX4	Mixed load, lateral force and camber on Mx	x	x	x	x	x	x			
q_{sx5}	QSX5	Load effect on Mx with lateral force and camber	x	x	x	x	x	x			
q_{sx6}	QSX6	B-factor of load with Mx	x	x	x	x	x	x			
q_{sx7}	QSX7	Camber with load on Mx	x	x	x	x	x	x			
q_{sx8}	QSX8	Lateral force with load on Mx	x	x	x	x	x	x			
q_{sx9}	QSX9	B-factor of lateral force with load on Mx	x	x	x	x	x	x			
q_{sx10}	QSX10	Vertical force with camber on Mx	x	x	x	x	x	x			
q_{sx11}	QSX11	B-factor of vertical force with camber on Mx	x	x	x	x	x	x			
q_{sx12}	QSX12	Camber squared induced overturning moment	x	x	x	x					
q_{sx13}	QSX13	Lateral force induced overturning moment	x	x	x	x					
q_{sx14}	QSX14	Lateral force induced overturning moment with camber	x	x	x	x					
p_{pmx1}	PPMX1	Influence of inflation pressure on overturning moment	x	x	x	x					

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1.5 Rolling resistance moment My

MF-Tyre 5.2, 6.0, MF-MCTyre 1.1:

$$M_y = -R_0 F_z \lambda_{My} \left\{ q_{sy1} + q_{sy2} \frac{F_x}{F_{z0}} + q_{sy3} \left| \frac{V_x}{V_{ref}} \right| + q_{sy4} \left(\frac{V_x}{V_{ref}} \right)^4 \right\}$$

MF-Tyre 6.1, 6.2:

$$M_y = -R_0 F_{z0} \lambda_{My} \left\{ q_{sy1} + q_{sy2} \frac{F_x}{F_{z0}} + q_{sy3} \left| \frac{V_x}{V_{ref}} \right| + q_{sy4} \left(\frac{V_x}{V_{ref}} \right)^4 + q_{sy5} \gamma^2 + q_{sy6} \frac{F_z}{F_{z0}} \gamma^2 \right\} \left\{ \left(\frac{F_z}{F_{z0}} \right)^{q_{sy7}} \left(\frac{p}{p_0} \right)^{q_{sy8}} \right\}$$

	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[SCALING_COEFFICIENTS]										
λ_{My}	LMY	Scale factor of rolling resistance torque	x	x	x	x	x	x	x	x	x
	[ROLLING_COEFFICIENTS]										
q_{sy1}	QSY1	Rolling resistance torque coefficient	x	x	x	x	x	x	x	x	x
q_{sy2}	QSY2	Rolling resistance torque depending on Fx	x	x	x	x	x	x	x	x	x
q_{sy3}	QSY3	Rolling resistance torque depending on speed	x	x	x	x	x	x	x	x	x
q_{sy4}	QSY4	Rolling resistance torque depending on speed ^4	x	x	x	x	x	x	x	x	x
q_{sy5}	QSY5	Rolling resistance torque depending on camber squared	x	x	x	x					
q_{sy6}	QSY6	Rolling resistance torque depending on load and camber squared	x	x	x	x					
q_{sy7}	QSY7	Rolling resistance torque coefficient load dependency	x	x	x	x					
q_{sy8}	QSY8	Rolling resistance torque coefficient pressure dependency	x	x	x	x					

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1.6 Self aligning moment Mz

MF-Tyre 5.2, MF-MCTyre 1.1: $M_z = -t \cdot \left(F_y - S_{Vy\kappa}\right) + M_{zr} + s \cdot F_x$

MF-Tyre 6.0, 6.1, 6.2: $M_z = -t \cdot F_{y0} \cdot G_{y\kappa 0} + M_{zr} + s \cdot F_x$

where $F_{y0} \cdot G_{y\kappa 0}$ is the side force with zero inclination angle ($\gamma = 0$) and zero turnslip ($\varphi_t = 0$)

MF-MCTyre 1.1: $\alpha_t = \alpha_M$ and $\alpha_r = \alpha_M + S_{Ht}$

MF-Tyre 5.2, 6.0, 6.1, 6.2: $\alpha_t = \alpha_M + S_{Ht}$ and $\alpha_r = \alpha_F + S_{Hy} + \dfrac{S_{Vy}}{K_{y\alpha}}$

$$S_{Ht} = q_{Hz1} + q_{Hz2} df_z + \left(q_{Hz3} + q_{Hz4} df_z\right) \cdot \gamma$$

combined slip:

$$\alpha_{t,eq} = \arctan \sqrt{\tan^2(\alpha_t) + \left(\frac{K_{x\kappa}}{K_{y\alpha}}\right)^2 \kappa^2} \operatorname{sgn}(\alpha_t)$$

$$\alpha_{r,eq} = \arctan \sqrt{\tan^2(\alpha_r) + \left(\frac{K_{x\kappa}}{K_{y\alpha}}\right)^2 \kappa^2} \operatorname{sgn}(\alpha_r)$$

$$s = \left(s_{sz1} + s_{sz2}\left(\frac{F_y}{F_{z0}}\right) + \left(s_{sz3} + s_{sz4} df_z\right)\gamma \right) R_0 \lambda_s$$

pure slip:

$$\alpha_{t,eq} = \alpha_t$$
$$\alpha_{r,eq} = \alpha_r$$
$$s = 0$$

pneumatic trail t:

$$t = D_t \cos\left[C_t \arctan\left\{B_t \alpha_{t,eq} - E_t\left(B_t \alpha_{t,eq} - \arctan\left(B_t \alpha_{t,eq}\right)\right)\right\}\right] \cdot \cos(\alpha_M)$$

$$B_t = \left(q_{Bz1} + q_{Bz2} df_z + q_{Bz3} df_z^2\right) \cdot \left(1 + q_{Bz4}\gamma + q_{Bz5}|\gamma|\right) \cdot \frac{\lambda_{Ky\alpha}}{\lambda_{\mu y}}$$

$$C_t = q_{Cz1}$$

MF-Tyre 5.2, 6.0, 6.1, 6.2:

$$D_t = F_z \cdot \left(q_{Dz1} + q_{Dz2} df_z\right) \cdot \left(1 - p_{pz1} dpi\right) \cdot \left(1 + q_{Dz3}\gamma + q_{Dz4}\gamma^2\right) \cdot \frac{R_0}{F_{z0}} \cdot \lambda_t \cdot \zeta_5$$

MF-MCTyre 1.1: $D_t = F_z \cdot \left(q_{Dz1} + q_{Dz2} df_z\right) \cdot \left(1 + q_{Dz3}|\gamma| + q_{Dz4}\gamma^2\right) \cdot \dfrac{R_0}{F_{z0}} \cdot \lambda_t \cdot \zeta_5$

$$E_t = \left(q_{Ez1} + q_{Ez2} df_z + q_{Ez3} df_z^2\right) \cdot \left(1 + \left(q_{Ez4} + q_{Ez5}\gamma\right) \cdot \left(\frac{2}{\pi}\right) \cdot \arctan\left(B_t C_t \alpha_t\right)\right)$$

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residual moment Mzr:

$$M_{zr} = D_r \cos\left[\zeta_7 \arctan\left(B_r \alpha_{r,eq}\right)\right]$$

$$D_r = F_z R_0 \lambda_{\mu y} \cos(\alpha_M) \begin{bmatrix} (q_{Dz6} + q_{Dz7} df_z) \cdot \lambda_r \cdot \zeta_2 + (q_{Dz8} + q_{Dz9} df_z) \cdot \gamma \cdot \lambda_{Kz\gamma} \cdot (1 + p_{pz2} dpi) \cdot \zeta_0 \\ + (q_{Dz10} + q_{Dz11} df_z) \cdot \gamma |\gamma| \cdot \lambda_{Kz\gamma} \cdot \zeta_0 \end{bmatrix} - \zeta_8 + 1$$

$$B_r = \left(q_{Bz9} \cdot \frac{\lambda_{Ky\alpha}}{\lambda_{\mu y}} + q_{Bz10} \cdot B_y \cdot C_y \right) \zeta_6$$

Turn slip:

$$\zeta_5 = \cos\left(\arctan\left(q_{Dt\varphi 1} R_0 \varphi_M\right)\right)$$

$$\zeta_6 = \cos\left(\arctan\left(q_{Br\varphi 1} R_0 \varphi_M\right)\right)$$

$$\zeta_7 = C_r = \tfrac{2}{\pi} \arccos\left(\frac{M_{z\varphi 90}}{|D_{r\varphi}|} \right)$$

$$M_{z\varphi 90} = M_{z\varphi\infty} \tfrac{2}{\pi} \arctan\left(q_{Cr\varphi 2} R_0 |\varphi_t|\right) \cdot G_{y\kappa}$$

$$M_{z\varphi\infty} = q_{Cr\varphi 1} |\mu_y| R_0 F_z \sqrt{\frac{F_z}{F'_{z0}}} \lambda_{M\varphi}$$

$$\zeta_8 = 1 + D_{r\varphi}$$

$$D_{r\varphi} = D_{Dr\varphi} \sin\left(C_{Dr\varphi} \arctan\left(B_{Dr\varphi} R_0 \varphi_M\right)\right)$$

$$B_{Dr\varphi} = \frac{K_{z\gamma r0}}{C_{Dr\varphi} D_{Dr\varphi} (1 - \varepsilon_\gamma)}$$

$$C_{Dr\varphi} = q_{Dr\varphi 1}$$

$$D_{Dr\varphi} = M_{z\varphi\infty} / \sin\left(0.5\pi C_{Dr\varphi}\right)$$

$$K_{z\gamma r0} = F_z R_0 \left(q_{Dz8} + q_{Dz9} df_z + (q_{Dz10} + q_{Dz11} df) \cdot |\gamma|\right) \lambda_{Kz\gamma}$$

when turn slip is not used: $\zeta_5, \zeta_6, \zeta_7, \zeta_8 = 1$

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	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[SCALING_COEFFICIENTS]										
λ_t	LTR	Scale factor of Peak of pneumatic trail	x	x	x	x	x	x	x	x	x
λ_r	LRES	Scale factor for offset of residual torque	x	x	x	x	x	x	x	x	x
λ_s	LS	Scale factor of Moment arm of Fx	x	x	x	x	x	x	x	x	x
$\lambda_{M\varphi}$	LMP	Scale factor of parking moment	x	x	x	x	x	x			
	[ALIGNING_COEFFICIENTS]										
q_{Bz1}	QBZ1	Trail slope factor for trail Bpt at Fznom	x	x	x	x	x	x	x	x	x
q_{Bz2}	QBZ2	Variation of slope Bpt with load	x	x	x	x	x	x	x	x	x
q_{Bz3}	QBZ3	Variation of slope Bpt with load squared	x	x	x	x	x	x	x	x	x
q_{Bz4}	QBZ4	Variation of slope Bpt with camber	x	x	x	x	x	x	x	x	x
q_{Bz5}	QBZ5	Variation of slope Bpt with absolute camber	x	x	x	x	x	x	x	x	x
q_{Bz9}	QBZ9	Slope factor Br of residual torque Mzr	x	x	x	x	x	x	x	x	x
q_{Bz10}	QBZ10	Slope factor Br of residual torque Mzr	x	x	x	x	x	x	x	x	x
q_{Cz1}	QCZ1	Shape factor Cpt for pneumatic trail	x	x	x	x	x	x	x	x	x
q_{Dz1}	QDZ1	Peak trail Dpt" = Dpt*(Fz/Fznom*R0)	x	x	x	x	x	x	x	x	x
q_{Dz2}	QDZ2	Variation of peak Dpt with load	x	x	x	x	x	x	x	x	x
q_{Dz3}	QDZ3	Variation of peak Dpt with camber	x	x	x	x	x	x	x	x	x
q_{Dz4}	QDZ4	Variation of peak Dpt with camber squared	x	x	x	x	x	x	x	x	x
q_{Dz6}	QDZ6	Peak residual torque Dmr = Dmr/(Fz*R0)	x	x	x	x	x	x	x	x	x
q_{Dz7}	QDZ7	Variation of peak factor Dmr with load	x	x	x	x	x	x	x	x	x
q_{Dz8}	QDZ8	Variation of peak factor Dmr with camber	x	x	x	x	x	x	x	x	x
q_{Dz9}	QDZ9	Variation of peak factor Dmr with camber and load	x	x	x	x	x	x	x	x	x
q_{Dz10}	QDZ10	Variation of peak factor Dmr with camber squared	x	x	x	x	x	x			x
q_{Dz11}	QDZ11	Variation of Dmr with camber squared and load	x	x	x	x	x	x			x
q_{Ez1}	QEZ1	Trail curvature Ept at Fznom	x	x	x	x	x	x	x	x	x
q_{Ez2}	QEZ2	Variation of curvature Ept with load	x	x	x	x	x	x	x	x	x
q_{Ez3}	QEZ3	Variation of curvature Ept with load squared	x	x	x	x	x	x	x	x	x
q_{Ez4}	QEZ4	Variation of curvature Ept with sign of Alpha-t	x	x	x	x	x	x	x	x	x
q_{Ez5}	QEZ5	Variation of Ept with camber and sign Alpha-t	x	x	x	x	x	x	x	x	x
q_{Hz1}	QHZ1	Trail horizontal shift Sht at Fznom	x	x	x	x	x	x	x	x	x
q_{Hz2}	QHZ2	Variation of shift Sht with load	x	x	x	x	x	x	x	x	x
q_{Hz3}	QHZ3	Variation of shift Sht with camber	x	x	x	x	x	x	x	x	x
q_{Hz4}	QHZ4	Variation of shift Sht with camber and load	x	x	x	x	x	x	x	x	x
s_{sz1}	SSZ1	Nominal value of s/R0: effect of Fx on Mz	x	x	x	x	x	x	x	x	x
s_{sz2}	SSZ2	Variation of distance s/R0 with Fy/Fznom	x	x	x	x	x	x	x	x	x
s_{sz3}	SSZ3	Variation of distance s/R0 with camber	x	x	x	x	x	x	x	x	x
s_{sz4}	SSZ4	Variation of distance s/R0 with load and camber	x	x	x	x	x	x	x	x	x
p_{pz1}	PPZ1	Linear pressure effect on pneumatic trail	x	x	x	x					
p_{pz2}	PPZ2	Influence of inflation pressure on residual aligning torque	x	x	x	x					
	[TURNSLIP_COEFFICIENTS]										
$q_{Dt\varphi1}$	QDTP1	Pneumatic trail reduction factor due to turn slip parameter	x	x	x	x	x	x			
$q_{Cr\varphi1}$	QCRP1	Turning moment at constant turning and zero forward speed parameter	x	x	x	x	x	x			
$q_{Cr\varphi2}$	QCRP2	Turn slip moment (at alpha=90deg) parameter for increase with spin	x	x	x	x	x	x			
$q_{Br\varphi1}$	QBRP1	Residual (spin) torque reduction factor parameter due to side slip	x	x	x	x	x	x			
$q_{Dr\varphi1}$	QDRP1	Turn slip moment peak magnitude parameter	x	x	x	x	x	x			

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2 Basic Tyre Properties

2.1 Loaded radius

Traditionally the tyre force and moment characteristics are defined in the tyre road contact point. The location of this point is defined by considering the tyre/wheel combination as an infinitely thin disk through the plane of symmetry of the tyre, as is shown in Figure 2.

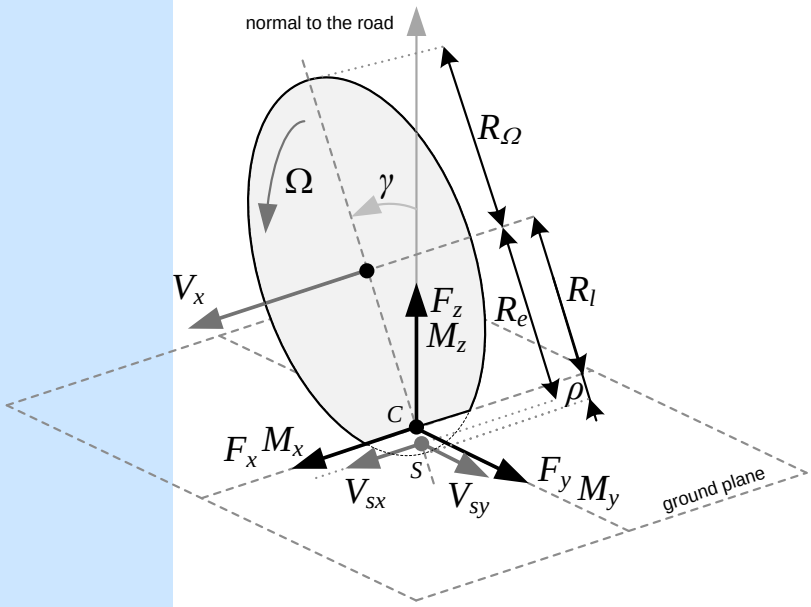

Figure 2: Forces, moments and kinematics of the tyre road contact (ISO sign convention) [7]

It is important to note that in practice the forces and moments are not measured at the tyre road contact point C, but at the wheel centre. In order to process the measurements and to calculate the forces and moments at the ground contact point, both the height of the wheel centre and inclination angle γ are required. Consequently, the distance from wheel centre to ground contact point, i.e. the loaded radius R_l, should be represented accurately in the tyre simulation model.

First centrifugal growth of the free tyre radius R_Ω is calculated using the following formula:

$$R_\Omega = R_0 \cdot \left(q_{re0} + q_{v1} \left(\frac{\Omega R_0}{V_0} \right)^2 \right)$$

Where R_0 equals the non-rolling free tyre radius, V_0 is the reference velocity, Ω the wheel rotational velocity and q_{re0} and q_{v1} are model parameters. The tyre deflection ρ_{zfr} for a free rolling tyre is the difference between the free tyre radius R_Ω and the loaded radius R_l:

$$\rho_{zfr} = \max\left(R_\Omega - R_l, 0\right)$$

Further, various effects are included in the calculation of the vertical force. First, an additional deflection caused by camber (q_{cam1}, q_{cam2}, q_{cam3}) is considered and combined into one tyre deflection ρ_z as is shown in the following equations, where γ equals the camber angle and rtw the reference tread width.

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$$\rho_{z\gamma} = \frac{\left(\left[q_{cam1} \cdot R_l + q_{cam2} \cdot R_l^2\right] \cdot \gamma\right)^2 \left(\frac{rtw}{8} \left|\tan(\gamma)\right|\right)}{\left(\left[q_{cam1} \cdot R_\Omega + q_{cam2} \cdot R_\Omega^2\right] \cdot \gamma\right)^2} - \left(q_{cam3} \cdot \rho_{zfr} \cdot |\gamma|\right)$$

$$\rho_z = \max\left(\rho_{zfr} + \rho_{z\gamma}, 0\right)$$

Secondly, all effects are combined in one correction term, containing a stiffness increase with velocity (q_{v2}), vertical sinking due to longitudinal forces (q_{Fcx}), vertical sinking due to lateral forces (q_{Fcy}, q_{Fcy2}) and the influence of the tyre inflation pressure (p_{Fz1}). Further, F_{z0} is the nominal load and *dpi* the non-dimensional inflation pressure increment

$$f_{corr} = \left(1 + q_{v2} \frac{R_0}{V_0} |\Omega| - \left(\frac{q_{Fcx} F_x}{F_{z0}}\right)^2 - \left(\left[\frac{\rho_z}{r_0}\right]^{q_{Fcy2}} \frac{q_{Fcy}\left(F_y - S_{Fy\gamma}\right)}{F_{z0}}\right)^2\right) \cdot \left(1 + p_{Fz1} \cdot dpi\right)$$

The shift $S_{Fy\gamma}$ describes the asymmetric effect for combinations of camber and lateral force (q_{Fys1}, q_{Fys2}, q_{Fys3}) that is defined as:

$$S_{Fy\gamma} = \left(q_{Fys1} + q_{Fys2} \frac{R_l}{R_\Omega} + q_{Fys3} \left(\frac{R_l}{R_\Omega}\right)^2\right) \cdot \gamma$$

Finally, all effects are considered when the vertical tyre force F_z is calculated using a quadratic force deflection characteristic (q_{Fz1}, q_{Fz2}) as is shown by the following formula:

$$F_z = f_{corr} \cdot \left(q_{Fz1} \frac{\rho_z}{R_0} + q_{Fz2} \left(\frac{\rho_z}{R_0}\right)^2\right) \cdot F_{z0}$$

For large camber angles (e.g. motorcycle tyres) an extended approach for calculating the vertical force can be used taking into account the contour of the tyre, see section 4.1.

When the tyre deflection becomes very large, the tyre belt and sidewalls may contact the rim. If this happens the vertical force will become suddenly very large due to the high (rim) stiffness felt at that moment. In MF-Swift a simple bottoming model is used that is shown in Figure 3 and explained below.

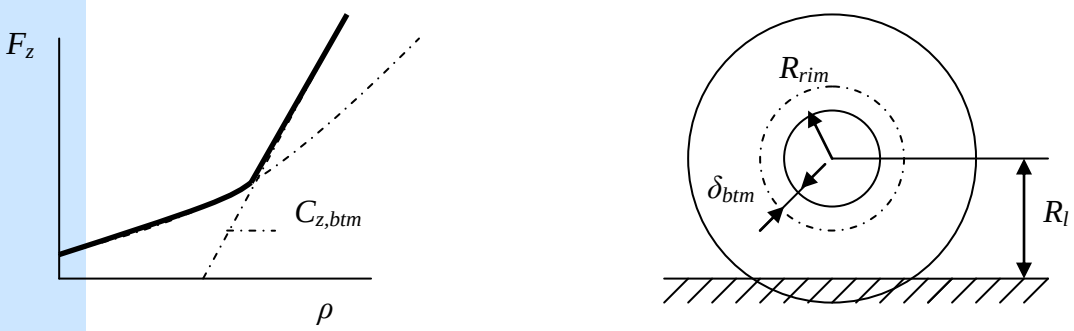

Figure 3: Bottoming model

Bottoming force $F_{z,btm}$ with bottoming offset δ_{btm} and large (rim) stiffness $C_{z,btm}$ (see Figure 3):

$$F_{z,btm} = C_{z,btm}\left(R_{rim} + \delta_{btm} - R_l\right)$$

Bottoming model:

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$$F_z = \max(F_z, F_{zbtm})$$

In the tyre property file the vertical stiffness C_{z0} at the nominal vertical load, nominal inflation pressure, no tangential forces and zero forward velocity is specified instead of q_{Fz1}. C_{z0} can be calculated as:

$$C_{z0} = \frac{F_{z0}}{R_0}\sqrt{q_{Fz1}^{\ 2} + 4q_{Fz2}} \ .$$

	Tyre property file			MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[DIMENSION]											
R_0	UNLOADED_RADIUS	Free tyre radius		x	x	x	x	x	x	x	x	x
R_{rim}	RIM_RADIUS	Nominal rim radius		x	x	x	x	x	x	x	x	x
	[VERTICAL]											
F_{z0}	FNOMIN	Nominal wheel load		x	x	x	x	x	x	x	x	x
C_{z0}	VERTICAL_STIFFNESS	Tyre vertical stiffness		x	x	x	x	x	x	x	x	x
q_{re0}	Q_RE0	Ratio of free tyre radius with nominal tyre radius		x	x	x	x	x	x		x	
q_{v1}	Q_V1	Tyre radius increase with speed		x	x	x	x	x	x		x	
q_{v2}	Q_V2	Vertical stiffness increase with speed		x	x	x	x	x	x		x	
q_{Fz2}	Q_FZ2	Quadratic term in load vs. deflection		x	x	x	x	x	x		x	
q_{Fcx}	Q_FCX	Longitudinal force influence on vertical stiffness		x	x	x	x	x	x		x	
q_{Fcy}	Q_FCY	Lateral force influence on vertical stiffness		x	x	x	x	x	x		x	
q_{Fcy2}	Q_FCY2	Explicit load dependency for including the lateral force influence on vertical stiffness		x	x							
q_{cam1}	Q_CAM1	Linear load dependent camber angle influence on vertical stiffness		x	x							
q_{cam2}	Q_CAM2	Quadratic load dependent camber angle influence on vertical stiffness		x	x							
q_{cam3}	Q_CAM3	Linear load and camber angle dependent reduction on vertical stiffness		x	x							
q_{Fys1}	Q_FYS1	Combined camber angle and side slip angle effect on vertical stiffness (constant)		x	x							
q_{Fys2}	Q_FYS2	Combined camber angle and side slip angle linear effect on vertical stiffness		x	x							
q_{Fys3}	Q_FYS3	Combined camber angle and side slip angle quadratic effect on vertical stiffness		x	x							
p_{Fz1}	PFZ1	Pressure effect on vertical stiffness		x	x	x	x					
δ_{btm}	BOTTOM_OFFST	Distance to rim when bottoming starts to occur		x	x	x	x	x	x		x	
$C_{z,btm}$	BOTTOM_STIFF	Vertical stiffness of bottomed tyre		x	x	x	x	x	x		x	

2.2 Effective rolling radius

In the expressions for the effective rolling radius and contact patch dimensions the vertical stiffness adapted for tyre inflation pressure is used:

$$C_z = C_{z0}(1 + p_{Fz1}dp_i)$$

The ratio of the forward velocity of the wheel centre V_x to the angular velocity of the free rolling tyre Ω_{fr} equals the effective rolling radius R_e. The following empirical formula is used:

$$R_e = R_\Omega - \frac{F_{z0}}{C_z}\left(D_{reff}\arctan\left(B_{reff}\frac{F_z}{F_{z0}}\right) + F_{reff}\frac{F_z}{F_{z0}}\right)$$

where D_{reff}, B_{reff} and F_{reff} are model parameters. The effective rolling radius defines the slip point S. Note that the location of S is different from the contact point C as is shown in Figure 2. Point S is used to calculate the sideslip angle α_F using the lateral slip velocity V_{sy}, see also section 1.1.

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	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[DIMENSION]										
R_0	UNLOADED_RADIUS	Free tyre radius	x	x	x	x	x	x	x	x	x
	[VERTICAL]										
F_{z0}	FNOMIN	Nominal wheel load	x	x	x	x	x	x	x	x	x
C_{z0}	VERTICAL_STIFFNESS	Tyre vertical stiffness	x	x	x	x	x	x	x	x	x
B_{reff}	BREFF	Low load stiffness e.r.r.	x	x	x	x	x	x	x	x	x
D_{reff}	DREFF	Peak value of e.r.r.	x	x	x	x	x	x	x	x	x
F_{reff}	FREFF	High load stiffness e.r.r.	x	x	x	x	x	x	x	x	x
p_{Fz1}	PFZ1	Pressure effect on vertical stiffness	x	x	x	x					

2.3 Contact patch dimensions

When modelling the tyre-road enveloping and relaxation behaviour the dimensions of the tyre contact patch are needed. The empirical expressions for half of the contact length *a* and half of the width *b* read:

$$a = R_0\left(q_{ra2}\frac{F_z}{C_z R_0} + q_{ra1}\sqrt{\frac{F_z}{C_z R_0}}\right) \approx R_0\left(q_{ra2}\frac{\rho}{R_0} + q_{ra1}\sqrt{\frac{\rho}{R_0}}\right)$$

$$b = w\left(q_{rb2}\frac{F_z}{C_z R_0} + q_{rb1}\left(\frac{F_z}{C_z R_0}\right)^{\frac{1}{3}}\right) \approx w\left(q_{rb2}\frac{\rho}{R_0} + q_{rb1}\left(\frac{\rho}{R_0}\right)^{\frac{1}{3}}\right)$$

where *w* is the nominal section width of the tyre. Since these expressions are functions of the tyre deflection *ρ*, the effect of changing the tyre inflation pressure is taken into account. Lowering the inflation pressure results in an increase in tyre deflection and thus an increase in contact length.

In older tyre models (6.0 & before) no influence of the tyre inflation pressure on the contact length is considered. The empirical expressions for half of the contact length *a* and half of the width *b* then read:

$$a = R_0\left(q_{a2}\frac{F_z}{F_{z0}} + q_{a1}\sqrt{\frac{F_z}{F_{z0}}}\right)$$

$$b = \frac{w}{2}$$

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	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[DIMENSION]										
R_0	UNLOADED_RADIUS	Free tyre radius	x	x	x	x	x	x	x	x	x
w	WIDTH	Nominal section width of the tyre	x	x	x	x	x	x	x	x	x
	[CONTACT_PATCH]										
q_{ra1}	Q_RA1	Square root term in contact length equation		x		x					
q_{ra2}	Q_RA2	Linear term in contact length equation		x		x					
q_{rb1}	Q_RB1	Root term in contact width equation		x		x					
q_{rb2}	Q_RB2	Linear term in contact width equation		x		x					
q_{a1}	Q_A1	Square root load term in contact length						x		x	
q_{a2}	Q_A2	Linear load term in contact length						x		x	

2.4 Relaxation behaviour

2.4.1 Empirical relaxation length

A tyre does not respond instantaneously to a change in sideslip angle, but a certain distance needs to be travelled before the steady-state levels of side force and moment are reached (similar for F_x). In the MF-Tyre model this phenomenon is typically modelled using a first-order differential equation with the relaxation length as 'distance' constant. In the MF-Swift model the dynamics are modelled using a rigid ring approach, residual stiffness and contact patch relaxation model. Ultimately these two different approaches could potentially result in simulating a different relaxation length for the same tyre, which obviously is not acceptable. Depending on the application there will be a need to be able to switch the model from a simple representation of the dynamics to a more elaborate but more time consuming variant. Besides the vertical force and the side slip angle, the inflation pressure is also considered in the model, because of its impact on the relaxation behaviour.

The solution for this combined set of requirements is to model the overall longitudinal and lateral tyre stiffness and calculate the required parameters from these expressions. The following expressions are used to describe the overall longitudinal C_x and lateral stiffness C_y of the tyre at ground contact:

$$C_x = C_{x0}\left(1 + p_{cfx1} df_z + p_{cfx2} df_z^2\right)\left(1 + p_{cfx3} dp_i\right)$$

$$C_y = C_{y0}\left(1 + p_{cfy1} df_z + p_{cfy2} df_z^2\right)\left(1 + p_{cfy3} dp_i\right)$$

where C_{x0} and C_{y0} are the longitudinal and lateral stiffness of the tyre at the nominal vertical force and inflation pressure. Using these stiffness and the longitudinal slip stiffness $K_{x\kappa}$ and cornering stiffness $K_{y\alpha}$ the relaxation lengths for longitudinal and sideslip are respectively:

$$\sigma_x = \frac{K_{x\kappa}}{C_x}$$

$$\sigma_y = \frac{K_{y\alpha}}{C_y}$$

In principle the stiffness C_x and C_y could be measured on a non-rolling tyre, but the preferred approach is to measure the cornering stiffness and lateral relaxation length in a transient test for a number of different inflation pressures and vertical loads. Subsequently the lateral stiffness equation

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can be fitted to these measurement points accordingly. Figure 4 gives an impression of the dependency of the lateral relaxation length on the vertical force and tyre inflation pressure for a standard passenger car tyre.

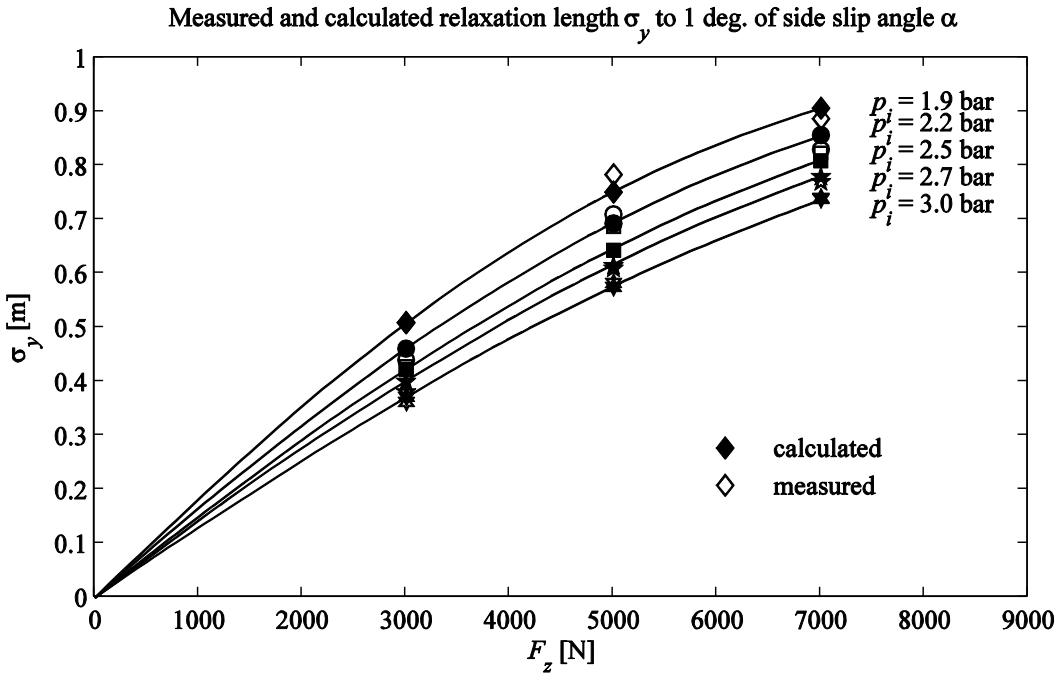

Figure 4: Lateral relaxation length as a function of vertical force and inflation pressure [7].

Next to the steady-state representation, three approaches with increasing complexity can be used to model the tyre transient behaviour and dynamics. The first two are explained in section 2.4.2 and 2.4.3, and the third in chapter 3 (MF-Swift).

2.4.2 Linear transients

The relaxation lengths determined in the previous paragraph are used in the next two differential equations to calculate the (transient) slip quantities.

$$\sigma_x \dot{\kappa} = -V_x \kappa - V_{sx}$$

$$\sigma_y \dot{\alpha}_F = -V_x \alpha_F + V_{sy}$$

MF-Tyre/MF-SWIFT 6.0, 6.1, 6.2:

$$\sigma_x = \frac{K_{x\kappa 0}}{C_x}$$

$$\sigma_y = \frac{K_{y\alpha 0}}{C_y}$$

Note: $K_{x\kappa 0}$ and $K_{y\alpha 0}$ are slip and cornering stiffness at $\kappa = 0$ and $\alpha = 0$.

MF-Tyre 5.2, SWIFT 1.2, MF-MCTyre 1.1:

$$\sigma_x = \left(p_{Tx1} + p_{Tx2} df_z\right) e^{-p_{Tx3} df_z} \lambda_{\sigma x} R_0 \frac{F_z}{F_{z0}}$$

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MF-MCTyre 1.1:

$$\sigma_{y\gamma} = 1 - p_{ky5}\gamma^2$$

$$p_{TyFzn} = p_{Ty3} + p_{Ky4}\gamma^2 F_{z0}^{'}$$

$$\sigma_y = p_{Ty1}\sin\left(p_{Ty2}\arctan\left(\frac{F_z}{p_{TyFzn}}\right)\right)\sigma_{y\gamma}R_0\lambda_{Fz0}\lambda_{\sigma y}$$

MF-Tyre 5.2, SWIFT 1.2:

$$\sigma_{y\gamma} = 1 - p_{ky3}|\gamma|$$

$$p_{TyFzn} = p_{Ty2}F_{z0}^{'}$$

$$\sigma_y = p_{Ty1}\sin\left(2\arctan\left(\frac{F_z}{p_{TyFzn}}\right)\right)\sigma_{y\gamma}R_0\lambda_{Fz0}\lambda_{\sigma y}$$

2.4.3 Nonlinear transients

In this approach the tyre-road contact is separated into two parts and can be considered as a series connection of a spring-damper with a relaxation system for the contact patch, see Figure 5.

$$k_{cy}\dot{\varepsilon}_y + C_{cy}\varepsilon_y = F_y$$

$$\sigma_c\dot{\alpha}_F = -V_x\alpha_F + V_{sy} + \dot{\varepsilon}_y$$

where σ_c is the contact patch relaxation length and ε_y is the lateral carcass deflection. The carcass stiffness C_{cy} is calculated as:

$$C_{cy} = \frac{K_{y\alpha}}{K_{y\alpha} - C_y a}C_y$$

A similar method is used in the longitudinal direction. The benefit of this approach is the resulting decreasing relaxation behaviour with increasing sideslip angle. Also the relaxation behaviour of the tyre on changes in vertical force is more accurately captured in this approach.

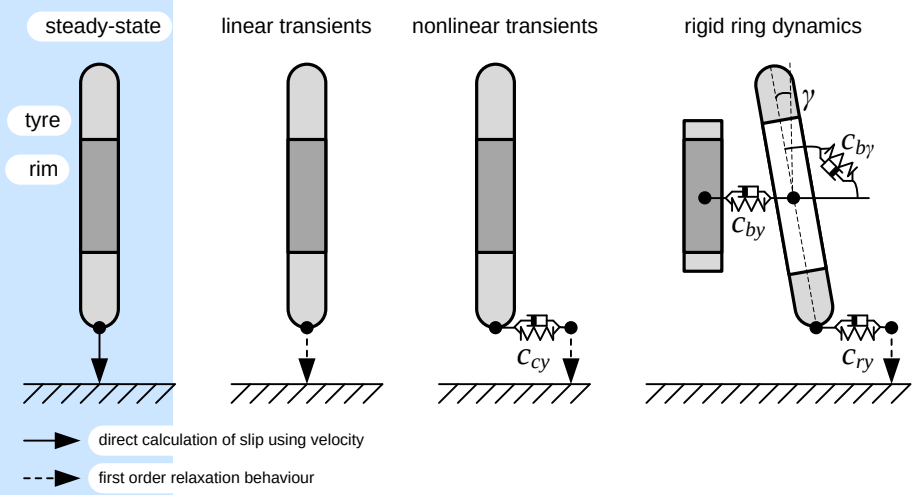

Figure 5: Schematic overview of different ways to model the contact transients/dynamics [7].

	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[STRUCTURAL]										
C_{x0}	LONGITUDINAL_STIFFNESS	Tyre overall longitudinal stiffness	x	x	x	x	x	x			
C_{y0}	LATERAL_STIFFNESS	Tyre overall lateral stiffness	x	x	x	x	x	x			
p_{cfx1}	PCFX1	Tyre overall longitudinal stiffness vertical deflection dependency linear term	x	x	x	x					
p_{cfx2}	PCFX2	Tyre overall longitudinal stiffness vertical deflection dependency quadratic term	x	x	x	x					
p_{cfx3}	PCFX3	Tyre overall longitudinal stiffness pressure dependency	x	x	x	x					
p_{cfy1}	PCFY1	Tyre overall lateral stiffness vertical deflection dependency linear term	x	x	x	x					
p_{cfy2}	PCFY2	Tyre overall lateral stiffness vertical deflection dependency quadratic term	x	x	x	x					
p_{cfy3}	PCFY3	Tyre overall lateral stiffness pressure dependency	x	x	x	x					
	[SCALING_COEFFICIENTS]										
$\lambda_{\sigma x}$	LSGKP	Scale factor of Relaxation length of Fx							x	x	x
$\lambda_{\sigma y}$	LSGAL	Scale factor of Relaxation length of Fy							x	x	x
	[LONGITUDINAL_COEFFICIENTS]										
p_{Tx1}	PTX1	Relaxation length SigKap0/Fz at Fznom							x	x	x
p_{Tx2}	PTX2	Variation of SigKap0/Fz with load							x	x	x
p_{Tx3}	PTX3	Variation of SigKap0/Fz with exponent of load							x	x	x
	[LATERAL_COEFFICIENTS]										
p_{Ty1}	PTY1	Peak value of relaxation length SigAlp0/R0							x	x	x
p_{Ty2}	PTY2	Value of Fz/Fznom where SigAlp0 is extreme							x	x	x
p_{Ty3}	PTY3	Value of Fz/Fznom where Sig_alpha is maximum									x

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3 MF-Swift

In this section the structure of the MF-Swift model is discussed. The important elements of the MF-Swift tyre model are:

- Magic Formula
- Contact patch slip model
- Rigid ring
- Obstacle enveloping model

The Magic Formula describes the steady-state tyre slip forces and moments with high accuracy, see Chapter 1. The other elements are discussed in the following sections.

3.1 Contact patch slip model

An elegant method to implement the load and slip dependent transient behaviour in the tyre model is including the carcass compliance, see section 2.4.3. This flexibility already accounts for a major part of the overall relaxation effect. This approach of elastically suspending the contact patch with respect to the wheel rim integrates very well with the introduction of the tyre dynamics by using the rigid ring approach. As can be seen in Figure 6, MF-Swift features a series connection of various springs: the belt is elastically suspended with respect to the rim and residual springs connect the contact patch with the belt. The forces generated by the Magic Formula are applied to the small contact patch mass and are transferred to the wheel axle through a series of springs, dampers and the belt mass.

In addition to the overall longitudinal and lateral stiffness, C_x and C_y respectively (see section 2.4), the overall yaw stiffness C_ψ is considered in MF-Swift and reads:

$$C_\psi = C_{\psi 0} \cdot \frac{F_z}{F_{z0}} \cdot \left(1 + p_{cmz1} dp_i\right)$$

where $C_{\psi 0}$ is the overall yaw stiffness at the nominal vertical force and inflation pressure.

As indicated in Figure 6 (right) the self aligning moment M_z in combination with yaw carcass stiffness c_ψ will result in a yaw deflection of both the tyre belt and the contact patch. Due to this deflection, the sideslip angle seen by the contact patch α_c is smaller than the sideslip angle of the wheel plane α.

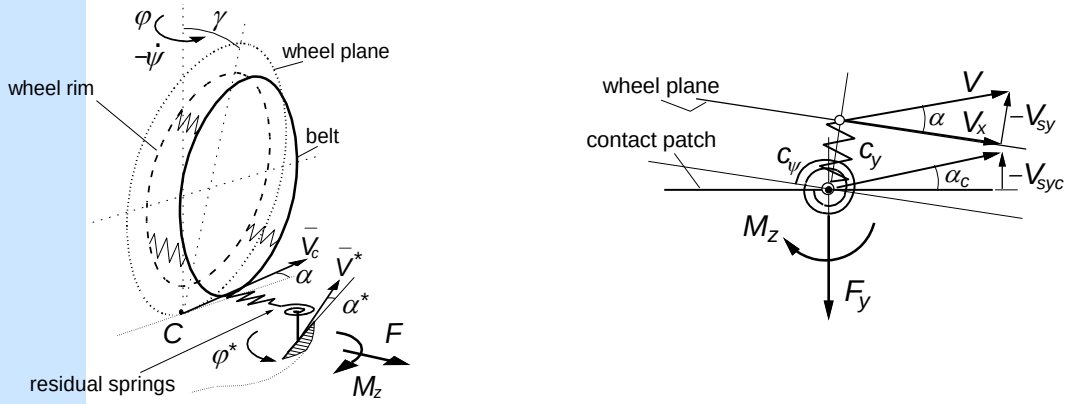

Figure 6: Schematic layout of the MF-Swift tyre model (left) and contact patch deflection detail (right).

The transient contact slip model used in MF-Swift accounts for the relaxation behaviour of the contact patch and enables starting from zero speed:

$$\sigma_c \dot{\alpha}_F^{'} + V_x \alpha_F^{'} = -V_{syc} - V_x \left(\frac{M_z}{C_\psi} \right)$$

with σ_c denoting the contact patch relaxation length, typically equal to half the contact length, and V_{syc} the lateral slip speed of the contact patch. To obtain the correct steady-state forces, the 'static' yaw deflection angle of the contact patch with respect to the wheel plane is used to compensate for the difference between the contact patch sideslip angle α_{Fc} and the wheel plane sideslip angle α that is used in the Magic Formula. The 'static' yaw deflection angle is calculated from the self aligning moment M_z and the yaw carcass stiffness c_ψ.

	Tyre property file			MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[STRUCTURAL]											
$C_{\psi 0}$	YAW_STIFFNESS	Tyre overall yaw stiffness		x	x	x	x	x	x			
p_{cmz1}	PCMZ1	Tyre overall yaw stiffness pressure dependency		x	x	x	x					

3.2 Rigid ring

The aim of the MF-Swift tyre model is to cover a higher frequency range that suits application for uneven roads and control systems. For these frequencies it is no longer allowed to consider the wheel and tyre as a single rigid body.

Scientific research [3] [4] [5] shows that if we restrict ourselves to a frequency range of approximately 60-100 Hz, the deformations of the tyre belt can be neglected. Consequently, the tyre belt is modelled as a rigid body, which is elastically suspended with respect to the rim for all six degrees of freedom. Residual springs are introduced between the tyre belt and the contact patch to ensure that the overall static stiffnesses are correct. Figure 7 shows the calculated primary tyre modes and the associated eigenfrequencies for a car tyre with the rigid ring model.

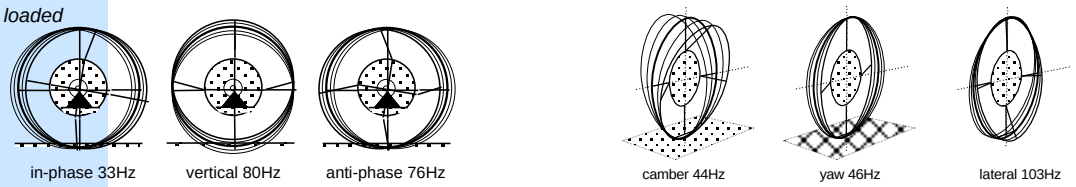

Figure 7: Calculated eigenfrequencies of a car tyre [6].

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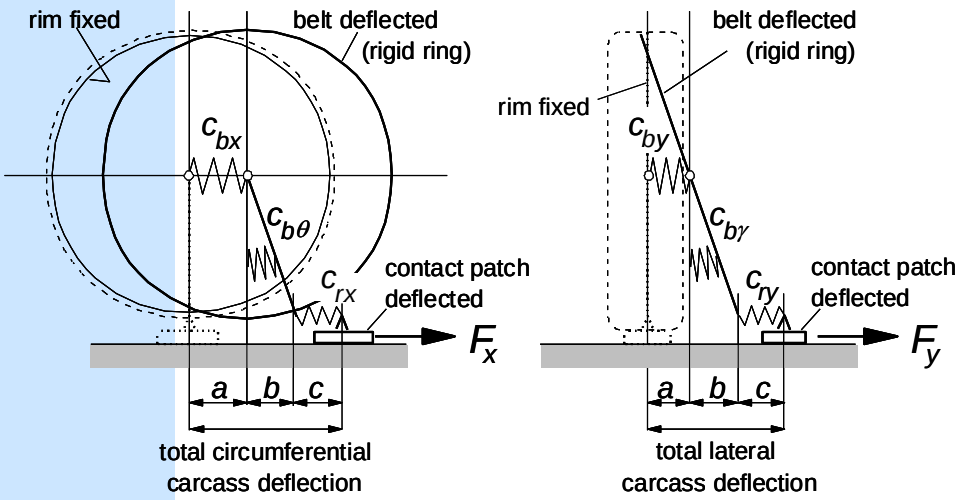

Figure 8: Rigid ring dynamics [6]

When including rigid ring dynamics, the spring-damper system is now subdivided into various components as shown in Figure 8. In this case the eigenfrequencies of the tyre belt determine the stiffness between rim and belt; still the overall stiffness is as specified in paragraph 2.4.1. This implies for example that the residual stiffnesses C_{rx},.C_{ry} & $C_{r\psi}$ are calculated from respectively:

$$\frac{1}{C_x} = \underbrace{\frac{1}{C_{bx}} + \frac{R_l^2}{C_{b\theta}} + \frac{1}{C_{rx}}}_{carcass} + \underbrace{\frac{a}{C_{f\kappa}}}_{contact\,patch}$$

$$\frac{1}{C_y} = \underbrace{\frac{1}{C_{by}} + \frac{R_l^2}{C_{b\gamma}} + \frac{1}{C_{ry}}}_{carcass} + \underbrace{\frac{a}{C_{f\alpha}}}_{contact\,patch}$$

$$\frac{1}{C_\psi} = \underbrace{\frac{1}{C_{b\psi}} + \frac{1}{C_{r\psi}}}_{carcass} + \underbrace{\frac{a/3}{C_{m\varphi}}}_{contact\,patch}$$

in which for instance C_{bx} and $C_{b\theta}$ are the longitudinal and rotational stiffness about the lateral axis between belt and rim, etc. The sidewall stiffnesses are calculated from the belt mass properties and the primary eigenfrequencies for the unloaded non-rolling tyre (subscript 0):

$$C_{bx0} = C_{bz0} = 4\pi^2 m_{belt} f_{long}^2$$

$$C_{by0} = 4\pi^2 m_{belt} f_{lat}^2$$

$$C_{b\theta 0} = 4\pi^2 I_{yy\,belt} f_{windup}^2$$

$$C_{b\gamma 0} = C_{b\psi 0} = 4\pi^2 I_{xx\,belt} f_{yaw}^2$$

The sidewall damping coefficients read as:

$$k_{bx} = k_{bz} = 4\zeta_{long}\pi f_{long} m_{belt}$$

$$k_{by} = 4\zeta_{lat}\pi f_{lat} m_{belt}$$

$$k_{b\theta} = 4\zeta_{windup}\pi f_{windup} I_{yy\,belt}$$

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$$k_{b\gamma} = 4\zeta_{yaw}\pi f_{yaw} I_{xxbelt}$$

where the symbol ζ is used for the dimensionless damping factors.

For the rolling tyre the sidewall stiffnesses may appear to increase with velocity. Further, the sidewall stiffnesses will be higher when the tyre inflation pressure is increased. Empirical relations have been developed to account for this effect:

$$C_{bx} = C_{bx0}\left(1+0.65dp_i\right)\left(1-q_{bvx}\sqrt{Q_V}\right)$$

$$C_{bz} = C_{bz0}\left(1+0.65dp_i\right)\left(1-q_{bvz}\sqrt{Q_V}\right), \qquad \text{with } q_{bvz} = q_{bvx}$$

$$C_{by} = C_{by0}\left(1+0.74dp_i\right)$$

$$C_{b\theta} = C_{b\theta 0}\left(1+0.49dp_i\right)\left(1-q_{bv\theta}\sqrt{Q_V}\right)$$

$$C_{b\gamma} = C_{b\gamma 0}\left(1+0.69dp_i\right)$$

$$C_{b\psi} = C_{b\psi 0}\left(1+0.69dp_i\right)$$

The variable Q_V is introduced here, which is a measure of the time rate of change of the loaded tyre deformation due to rolling:

$$Q_V = \frac{|\Omega|}{V_0}\sqrt{x_{rb}^2 + z_{rb}^2}$$

Note: The eigenfrequencies of the loaded and rolling tyre are different from the frequencies *f* defined in the tyre property file.

To calculate the residual stiffness $C_{r\psi}$ the moment stiffness against spin in the contact patch $C_{m\varphi}$ has to be known. $C_{m\varphi}$ is calculated using the following formulae:

$$K_{z\gamma r0} = F_z R_0\left(q_{dz8} + q_{dz9}df_z\right)\lambda_{kz\gamma}$$

$$\varepsilon_\gamma = p_{ecp1}\left(1+p_{ecp2}df_z\right)$$

$$C_{m\varphi} = \frac{R_0 K_{z\gamma r0}}{1-\varepsilon_\gamma}$$

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	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[DIMENSION]										
R_0	UNLOADED_RADIUS	Free tyre radius	x	x	x	x	x	x	x	x	x
	[INERTIA]										
m_{tyre}	MASS	Tyre mass	x	x	x	x	x	x		x	
I_{XXtyre}	IXX	Tyre diametral moment of inertia	x	x	x	x	x	x			
I_{YYtyre}	IYY	Tyre polar moment of inertia	x	x	x	x	x	x			
m_{belt}	BELT_MASS	Belt mass		x		x		x			
I_{XXbelt}	BELT_IXX	Belt diametral moment of inertia		x		x		x			
I_{YYbelt}	BELT_IYY	Belt polar moment of inertia		x		x		x			
g	GRAVITY	Gravity acting on belt in Z direction		x		x		x			
	[STRUCTURAL]										
f_{long}	FREQ_LONG	Undamped frequency fore/aft and vertical mode		x		x		x			
f_{lat}	FREQ_LAT	Undamped frequency lateral mode		x		x		x			
f_{yaw}	FREQ_YAW	Undamped frequency yaw and camber mode		x		x		x			
f_{windup}	FREQ_WINDUP	Undamped frequency wind-up mode		x		x		x			
ζ_{long}	DAMP_LONG	Dimensionless damping fore/aft and vertical mode		x		x		x			
ζ_{lat}	DAMP_LAT	Dimensionless damping lateral mode		x		x		x			
ζ_{yaw}	DAMP_YAW	Dimensionless damping yaw and camber mode		x		x		x			
ζ_{windup}	DAMP_WINDUP	Dimensionless damping wind-up mode		x		x		x			
ζ_{res}	DAMP_RESIDUAL	Residual damping (proportional to stiffness)	x	x	x	x	x	x			
q_{bvx}	Q_BVX	Load and speed influence on in-plane translation stiffness		x		x		x		x	
$q_{bv\theta}$	Q_BVT	Load and speed influence on in-plane rotation stiffness		x		x		x		x	
	[SCALING_COEFFICIENTS]										
$\lambda_{Kz\gamma}$	LKZC	Scale factor of camber moment stiffness	x	x	x	x	x	x			
	[ALIGNING_COEFFICIENTS]										
q_{Dz8}	QDZ8	Variation of peak factor Dmr with camber	x	x	x	x	x	x	x	x	x
q_{Dz9}	QDZ9	Variation of peak factor Dmr with camber and load	x	x	x	x	x	x	x	x	x
	[TURNSLIP_COEFFICIENTS]										
$p_{\varepsilon\gamma\varphi 1}$	PECP1	Camber w.r.t. spin reduction factor parameter in camber stiffness	x	x	x	x	x	x			
$p_{\varepsilon\gamma\varphi 2}$	PECP2	Camber w.r.t. spin reduction factor varying with load parameter in camber stiffness	x	x	x	x	x	x			

3.3 Obstacle enveloping model

The MF-Swift model should be able to meet the simulation targets for (short wavelength) road obstacles. An important restriction of the rigid ring model is that the contact between tyre and road is formed by a single point. In this point the tyre slip is calculated and the Magic Formula forces and moment are applied. Short wavelength obstacles give rise to nonlinear behaviour of the horizontal F_H and vertical F_V tyre forces and to changes of the effective rolling radius r_e, as illustrated in Figure 9.

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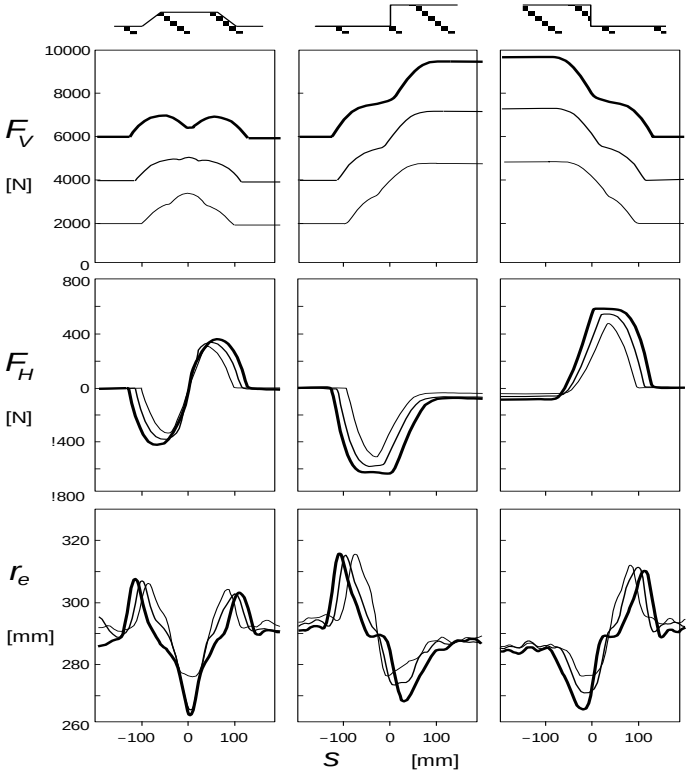

Figure 9: Measured forces and change of the effective rolling radius while driving over various obstacles with fixed axle height at low speed. Results are shown for three different initial vertical loads [6].

To incorporate these effects in the MF-Swift model the concept of the effective road surface is used. The idea behind this effective input is that it is assumed that the quasi-static response of a tyre model with a single-point tyre-road interface on the effective road surface is similar to the quasi-static response of the real tyre on the actual road surface. Furthermore, the assumption is made that the tyre contact zone dynamically deforms in the same way as it does quasi-statically and that, consequently, local dynamic effects in the contact area can be neglected; the rigid ring takes care of the dynamics.

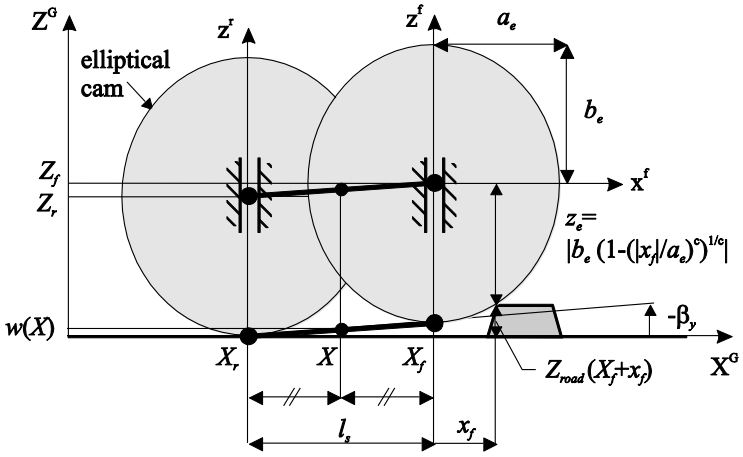

Figure 10: The tandem cam configuration, generating the effective road height and slope [5].

A technique using non-deformable elliptical cams which touch the actual road surface has been introduced to generate the effective road surface for arbitrary 3D road unevenness, see Figure 10, Figure 11 and Figure 12. Basically in the contact area a grid of elliptical cams is used to sense the road undulations. Then we can calculate:

- The vertical position of the effective road plane by taking the average of the z-coordinates across the width for both the front and rear row of cams.
- The slope of the effective road in the driving direction by calculating the average angle across the width of the tyre contact patch for the front and rear row of cams.
- The slope transverse to the driving direction (road camber/banking) by calculating the average angle across the length of the tyre contact patch for the left and right row of cams.

Motorcycle

For motorcycles two additional parameters are introduced describing an empirical relation. Parameter C_1 realizes attenuation of the vertical position (effective plane height) proportional to the difference in height of leading and trailing cams. The slope (effective plane angle) is adjusted with the same quantity, and parameter C_2 allows further attenuation of the effective plane angle to match the longitudinal force with more accuracy. The calculation then is:

- The effective plane height is calculated as described above, but additionally the factor $C_1|Z_1-Z_2|$ is subtracted, where Z_1 and Z_2 are the average heights of the trailing and leading cams respectively.
- The average slope of the effective road in the driving direction is calculated as described above, but is multiplied additionally with the reduction factor $(1-C_1)(1-C_2)$.

Further, note that the motorcycle contour contact is used on the effective road surface determined by the enveloping model. More details on the motorcycle contour can be found in paragraph 4.1.

The dimensions of the cams are defined by the ellipse parameters, see Figure 10. In terms of the coordinates x_e and z_e the ellipse equation reads:

$$\left(\frac{x_e}{a_e}\right)^{c_e}+\left(\frac{y_e}{b_e}\right)^{c_e}=1,$$

with

$$a_e = l_e R_0$$

$$b_e = h_e R_0$$

The effective road height and slope is accessed by using tandem cams at distance l_s, with

$$l_s = 2 p_{ls} a$$

It can be noted that the elliptical cams are only required on the outside contour of the contact patch. Furthermore, the contact patch dimensions (especially the contact length) change as a function of vertical load. This interaction is schematically shown in Figure 11.

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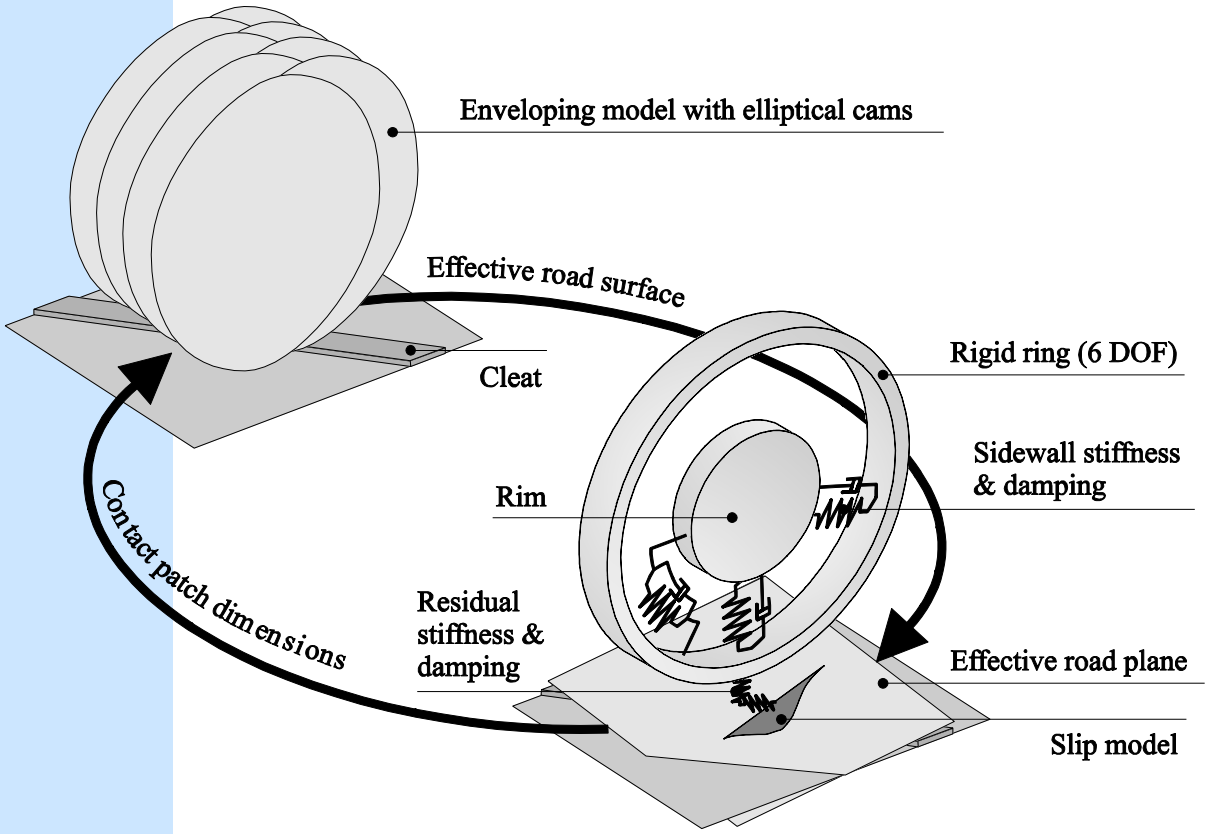

Figure 11: Schematic view of the MF-Swift model using the enveloping model with elliptical cams [5].

The shape of the elliptical cam is assessed by fitting model responses to measurements of the tyre rolling at very low velocity over obstacles.

In order to achieve higher accuracy at short wave length obstacles more parallel tandems (multi-track) and intermediate cams at the side edges can be used. In Figure 12 an example of 6 parallel cams ($N_{e,w}$) in the front & rear row and 5 successive cams at both sides ($N_{e,l}$) can be seen.

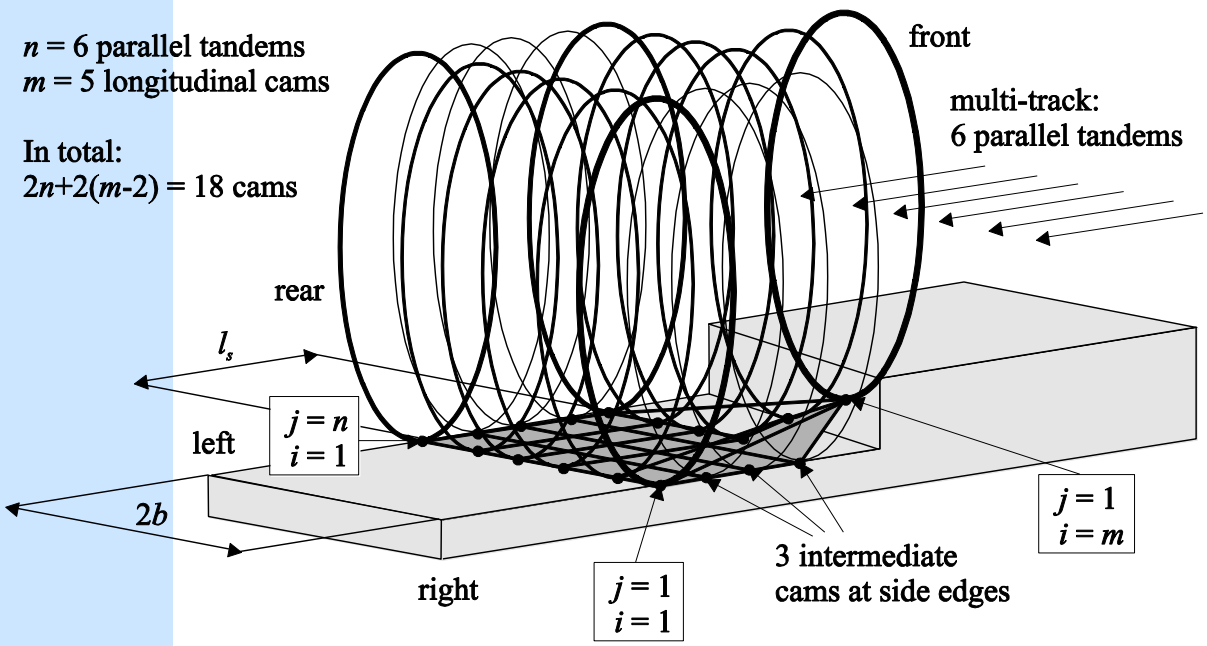

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Figure 12: Multiple cams to obtain higher accuracy. Example of '6x5' cams [5].

Research has shown that the shape of the elliptical cams does not change with tyre vertical load and inflation pressure, and that the tyre stiffness and contact length change cause the main effect.

	Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
	[CONTACT_PATCH]										
p_{ls}	ELLIPS_SHIFT	Scaling of distance between front and rear ellipsoid		x		x		x		x	
l_e	ELLIPS_LENGTH	Semimajor axis of ellipsoid		x		x		x		x	
h_e	ELLIPS_HEIGHT	Semiminor axis of ellipsoid		x		x		x		x	
c_e	ELLIPS_ORDER	Order of ellipsoid		x		x		x		x	
$N_{e,w}$	ELLIPS_NWIDTH	Number of parallel ellipsoids		x		x		x		x	
$N_{e,l}$	ELLIPS_NLENGTH	Number of ellipsoids at sides of contact patch		x		x		x		x	
C_1	ENV_C1	Effective height attenuation		x							
C_2	ENV_C2	Effective plane angle attenuation		x							

Note: The parameters ELLIPS_NWIDTH and ELLIPS_NLENGTH are simulation settings that can be set by the simulation engineer, see also section 4.4.

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4 Miscellaneous

4.1 Motorcycle Contour Ellipse (MF-Tyre/MF-Swift 6.1 and 6.2)

The motorcycle contour ellipse is the contact ellipse of a motorcycle tyre. It is the ellipse over which the motorcycle tyre rolls, as explained in the figure below.

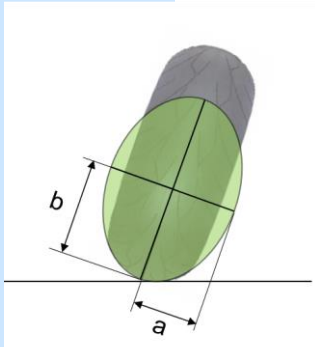

Figure 13: Motorcycle contour ellipse

The dimension of the ellipse can be set using the dimensionless parameters MCE_a and MCE_b according to the following formulas, where *a* and *b* here indicate the ellipse dimensions as shown in the figure above and *w* is the nominal section width of the tyre:

$$MCE_a = \frac{a}{w}$$

$$MCE_b = \frac{b}{w}$$

From release 6.2, the motorcycle contour contact is used when activating the Motorcycle Tyres Contact method, or the Contact Method for 2D and 3D roads if the contour parameters are nonzero. In the Contact Method for 2D and 3D roads cases, the motorcycle contour contacts the effective road surface instead of the real road surface.

Note: In the MF-Tyre 6.1 tyre model, the motorcycle contour contact is only supported when activating the smooth road contact, circular cross section (motorcycle tyres) contact method. Thus not in combination with the Contact Method for 2D and 3D roads.

4.2 Gyroscopic moment

For linear transients only, the gyroscopic moment M_{gyr} is calculated as:

$$M_{gyr} = q_{Tz1}\lambda_{gyr} m_{belt} V_r \dot{\alpha}_F \cos\left[\zeta_7 \arctan\left(B_r \alpha_{r,eq}\right)\right]$$

This moment is added to the aligning moment M_z.

4.3 Vertical stiffness and damping tyre

The vertical stiffness and damping of the tyre has been modelled as a one or two spring-damper system, which is dependent on the used tyre model, see figure below.

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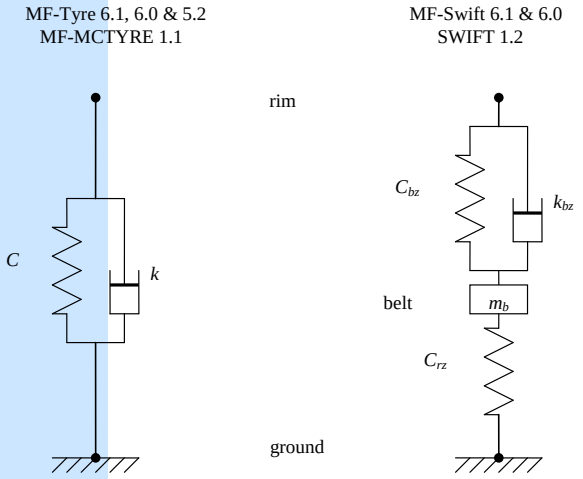

Figure 14: Simplified representation of vertical dynamics in MF-Tyre (left) and MF-Swift (right).

with C = vertical stiffness tyre (parameter VERTICAL_STIFFNESS C_{z0})

k = vertical damping tyre parameter VERTICAL_DAMPING k_{z0})

C_{bz} = vertical stiffness belt

k_{bz} = vertical damping belt (parameter DAMP_LONG ζ_{long})

m_b = mass belt (parameter BELT_MASS or MBELT m_{belt})

C_{rz} = residual vertical stiffness

	Tyre property file			**MF-Tyre 6.2**	**MF-Swift 6.2**	**MF-Tyre 6.1**	**MF-Swift 6.1**	**MF-Tyre 6.0**	**MF-Swift 6.0**	**MF-Tyre 5.2**	**SWIFT 1.2**	**MF-MCTyre 1.1**
	[DIMENSION]											
w	WIDTH	Nominal section width of the tyre		x	x	x	x	x	x	x	x	x
	[INERTIA]											
m_{belt}	BELT_MASS	Belt mass			x		x		x			
	[VERTICAL]											
C_{z0}	VERTICAL_STIFFNESS	Tyre vertical stiffness		x	x	x	x	x	x	x	x	x
k_{z0}	VERTICAL_DAMPING	Tyre vertical damping		x	x	x	x	x	x	x	x	x
MCE_a	MC_CONTOUR_A	Motorcycle contour ellips A		x		x						
MCE_b	MC_CONTOUR_B	Motorcycle contour ellips B		x		x						
	[STRUCTURAL]											
ζ_{long}	DAMP_LONG	Dimensionless damping fore/aft and vertical mode			x		x		x			
	[SCALING_COEFFICIENTS]											
λ_{gyr}	LGYR	Scale factor gyroscopic moment								x	x	x
	[ALIGNING_COEFFICIENTS]											
q_{Tz1}	QTZ1	Gyroscopic torque constant								x	x	x
m_{belt}	MBELT	Belt mass of the wheel								x	x	x

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4.4 Road surface simulation settings

Next to the tyre model parameters that describe the tyre characteristics, the tyre property file also contains some parameters that are used to change the settings of the tyre model for a specific simulation task. In general these parameters are to be set/modified by the engineer who performs the simulations. In the table below an overview of these parameters is given. For a more detailed description see the MF-Tyre and MF-Swift Help that is provided with the tyre model.

Parameter	Description
ROAD_DIRECTION	Direction of travelled distance to be used with road contact for 2D roads.
ROAD_INCREMENT	Increment in road sampling, typically equal to sample interval of measured road profile.
ELLIPS_MAX_STEP	Maximum height of road step. Specifies up to which height on the ellipse contour of the enveloping model contact points are considered. To be used with road contact for 3D roads.

Note: The simulation engineer can further influence the number of contact points that are evaluated by setting the parameters ELLIPS_NWIDTH and ELLIPS_NLENGTH, see section 3.3.

Tyre property file		MF-Tyre 6.2	MF-Swift 6.2	MF-Tyre 6.1	MF-Swift 6.1	MF-Tyre 6.0	MF-Swift 6.0	MF-Tyre 5.2	SWIFT 1.2	MF-MCTyre 1.1
[MODEL]										
ROAD_INCREMENT	Increment in road sampling		x		x		x		x	
ROAD_DIRECTION	Direction of travelled distance		x		x		x		x	
[CONTACT_PATCH]										
ELLIPS_MAX_STEP	Maximum height of road step		x		x		x		x	

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5 References

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